

Checking the Model, Seeing the Season

Module 5 · Residual diagnostics and seasonality

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Where we are: the last knob on the board

$$y_t = \alpha + \delta t + \sum_{j=1}^p \phi_j y_{t-j} + \sum_{l=1}^q \theta_l \epsilon_{t-l} + \epsilon_t$$

Term	L3	L4	L5
$\alpha, \delta t$	on	on	on
$\phi_j y_{t-j}$, all p	on	on	on
$\theta_l \epsilon_{t-l}$, all q	on	on	on
Seasonal block — the same structure at lag s	off	off	turning on today
ϵ_t	on	on	on

Two halves, one picture. **Part A:** learn to *check* a model — residuals are the final jury. **Part B:** the structure Module 4’s residuals kept pointing at — what seasonality is, and three ways to treat it.

By the end of today every slider on the mixing board is on.

Last time ended on a spike. Reproduce it.

```

1 library(forecast)
2 csv <- read.csv("../..data/UNRATE.csv") # cached FRED pull
3 csv <- subset(csv, observation_date >= "1960-01-01" &
4               observation_date <= "2019-12-01")
5 unrate <- ts(csv$UNRATE, start = c(1960, 1), frequency = 12)
6 d_unrate <- diff(unrate)
7
8 fit_l4 <- Arima(d_unrate, order = c(1, 0, 2), # Module 4's pick,
9               include.mean = FALSE) # same zero-mean policy
10 e_hat <- residuals(fit_l4)
11 ggAcf(e_hat, lag.max = 36)
12 acf(e_hat, lag.max = 36, plot = FALSE)$acf[c(13, 25, 37)]

```

$\widehat{\rho}_e(12)$	-0.145
------------------------	---------------

$\widehat{\rho}_e(24)$	-0.125
------------------------	---------------

$\widehat{\rho}_e(36)$	-0.130
------------------------	---------------

band, $1.96/\sqrt{T}$ ($\approx 2/\sqrt{T}$)	± 0.073
--	-------------------------------

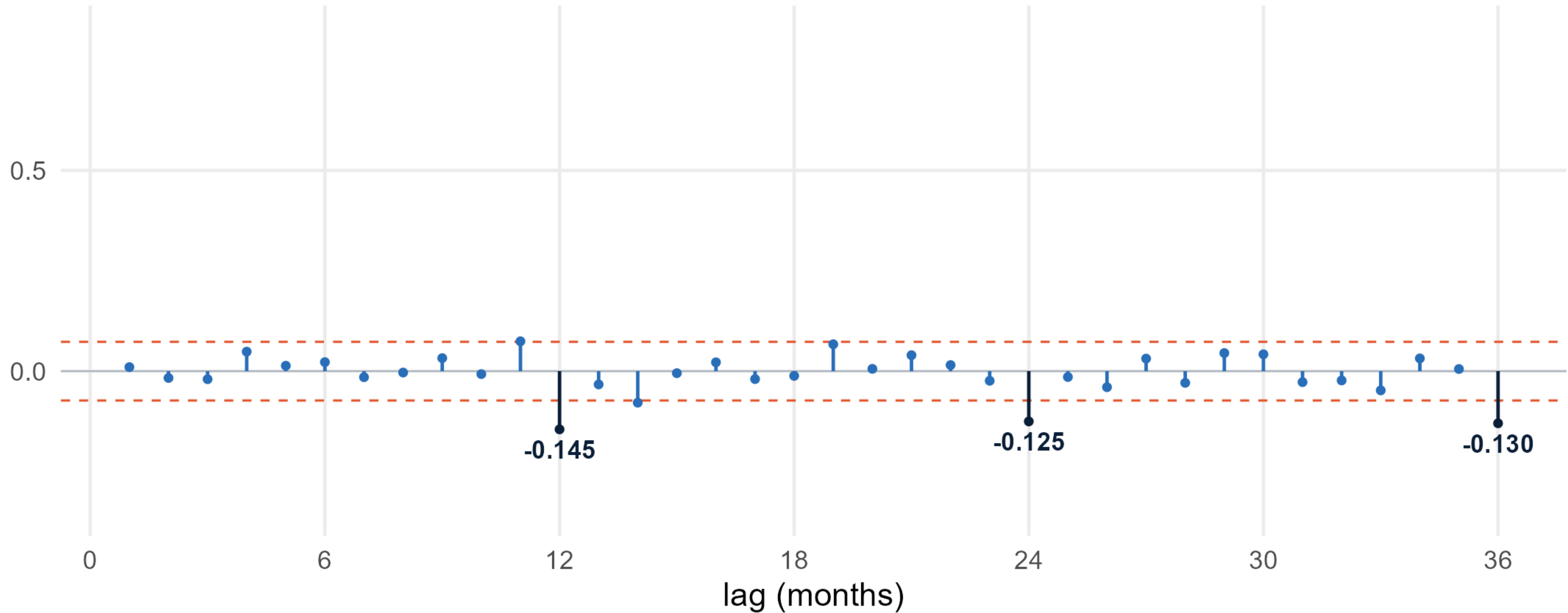
Same window as Module 4: 1960-01 to 2019-12, $T = 719$ differences. The low lags are quiet. The spikes every twelve months survive.

What would you need to see to sign off on this model?

Cached: the Module 4 residual ACF

Residual ACF: ARMA(1,2) on diff(UNRATE), 1960-2019

the Module 4 working model; dashed band = $\pm 1.96/\sqrt{T}$, $T = 719$



The principle: residuals should behave like innovations

- ϵ_t is the **innovation** — the DGP's shock. Unobserved.
- $e_t = y_t - \hat{y}_t$ is the **residual** — what the fitted model leaves. Observed.
- If the model is right, e_t should *behave like* $\epsilon_t \sim \text{WN}(0, \sigma^2)$. Not *is* white noise — no finite sample is — but *could plausibly be*.
- If it does not, there is nothing to fix in the data. There is something to fix in the model.

Four checks, one ranked line:

serial correlation → **heteroskedasticity** → **zero mean** → **normality**

The first is the whole of today. The second is a preview of Module 14 — eyeball it for now. The last two are cheap, and almost never the real problem.

Serial correlation: the definition, then the rule

Term	Meaning	Belongs to
Autocorrelation (dependence, memory, persistence)	A property of a series, measured by its ACF — any series, residuals included	the data
Serial correlation	The diagnostic verdict that a fitted model’s residuals e_t show autocorrelation	the model , relative to the data
Residual seasonality	Seasonal autocorrelation left in a published seasonally adjusted series	the adjustment procedure — itself a model

“Serial correlation is not a property of the data — it is a property of your model relative to the data. It means your model is wrong.”

Enders, Hamilton, and Hyndman & Athanasopoulos use autocorrelation and serial correlation interchangeably, and so does most applied practice. Recognize that usage in the wild; the course’s line is deliberate.



Neither is a property of the data.

Reading residuals against the Module 3 table

Residual ACF pattern	First candidate diagnosis — a proposal, not a verdict
Mostly inside $\pm 2/\sqrt{T}$, no systematic pattern	no visible evidence of remaining serial correlation
Spike at lag 1 only	an omitted short-run AR or MA term
Geometric decay from lag 1	omitted low-order AR dynamics
Cutoff after a few lags	omitted low-order MA dynamics
Spikes at $s, 2s, 3s$ (here 12, 24, 36)	omitted dependence at the seasonal frequency — today's transition
Large correlations that persist, not decay	insufficient differencing; reassess the transformation

Same table as Module 3, different question: there it told you what to *fit*; here, what your fit *missed*. Propose the next candidate, refit, re-check — a diagnosis earns its place only if the revised fit removes the pattern without creating a new one.

Misspecify and watch: predict the residual ACF before every move

Pick a DGP you know — AR(2), MA(2), or the two glued together — fit an ARMA(p, q), read the residual ACF and the Ljung-Box verdict. Predict before every move.

THE DATA (DGP)

AR(2)

MA(2)

ARMA(2,2)

Reseed

draw 1

YOUR MODEL

AR order $p =$



2

MA order $q =$

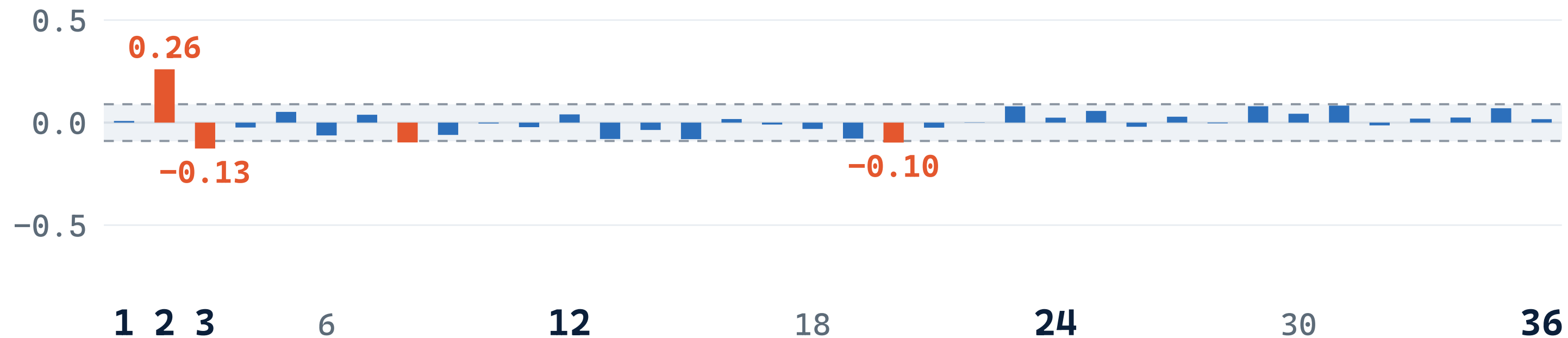


0

Reset

residual ACF $\hat{\rho}_e(k)$ by lag k

$\pm 2/\sqrt{n} = \pm 0.090$ ($n = 498$)



Fitted: AR(2) by CSS · $\hat{\mu} = -0.03$, $\hat{\phi}_1 = 0.86$, $\hat{\phi}_2 = 0.04$ · $n = 498$ · exact OLS · DGP ARMA(2,2), draw 1

Ljung-Box $Q(24) = 74.08$ · $h = 24$, $\text{fitdf} = 2$ · χ^2_{22} · $p = < 0.001$ **reject at 5%: serial correlation**

Residuals show serial correlation, largest at lag 2 ($\hat{\rho}_e = 0.26$) → this fit is missing structure.

Why (DGP known here, not in practice): missing MA structure — AR(2) truncates the $AR(\infty)$ form, leaving MA structure in e_t at lags 1–2.

Fits here are conditional sum of squares; R's `Arima()` uses maximum likelihood — the residual story is the same.

Three lessons from the simulation

1. **The diagnostic tools work.** When the residuals carry real structure, Ljung-Box flags it without knowing the DGP, and the residual ACF shape proposes what to add — a candidate, not a verdict.
2. **The fit will not volunteer the problem.** Fit the one-block-short model with `Arima()` and it prints coefficients, standard errors, an AIC, a log-likelihood. None of them says “short”. Stop at the IC winner without checking residuals and you own a confidently wrong model. Module 1’s spurious-regression Monte Carlo, in a new setting: the printout lies, and only a dependence-aware check catches it.
3. **The structure came from the model.** The DGP was perfectly well behaved. The serial correlation in the residuals was *generated by the fit*. The correct orders leave white noise — and extra orders leave white noise too, at a price diagnostics cannot see and only parsimony can. Change the fit and the structure is gone; the data never moved.

Serial correlation is a model report card, not a disease the data has.

Ljung-Box, on one slide

$$Q(h) = T(T+2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{T-k} \stackrel{a}{\sim} \chi_{h-\text{fitdf}}^2$$

- $\hat{\rho}_k$ is the sample autocorrelation of the **residuals** at lag k ; a joint test that the first h of them are all zero.
- **fitdf** = $p + q$ for a non-seasonal ARMA fit, $p + q + P + Q_s$ for a SARIMA fit, 0 for a raw series with no model. You spent those degrees of freedom estimating; the test charges for them.
- **Choose h before you look**: $h = 10$ for non-seasonal residuals, $h = 24$ for monthly seasonal residuals. Report both on monthly data. Hunting across h until one rejects is p -hacking with a chi-square.
- **Large p -value is good news**. In regression you want to reject. In diagnostics you want to *fail* to reject: no structure found.

```
1 Box.test(e_hat, lag = 10, type = "Ljung-Box", fitdf = 3)    # p + q = 1 + 2
2 Box.test(e_hat, lag = 24, type = "Ljung-Box", fitdf = 3)
3 checkresiduals(fit_14)                                     # the all-in-one from forecast
```

Verdict on the Module 4 fit

```
1 Box.test(e_hat, lag = 10, type = "Ljung-Box", fitdf = 3)
2 Box.test(e_hat, lag = 24, type = "Ljung-Box", fitdf = 3)
3 checkresiduals(fit_14)
```

$h = 10: Q = 3.76, \chi^2_7$ **p = 0.807 · pass**

$h = 24: Q = 46.1, \chi^2_{21}$ **p = 0.0012 · fail**

The two horizons disagree, and the disagreement is the lesson. Over ten lags these residuals are white noise. Widen the window to lag 24 and the joint statistic sees what the eye saw: a test that never looks at lag 12 cannot reject because of lag 12.

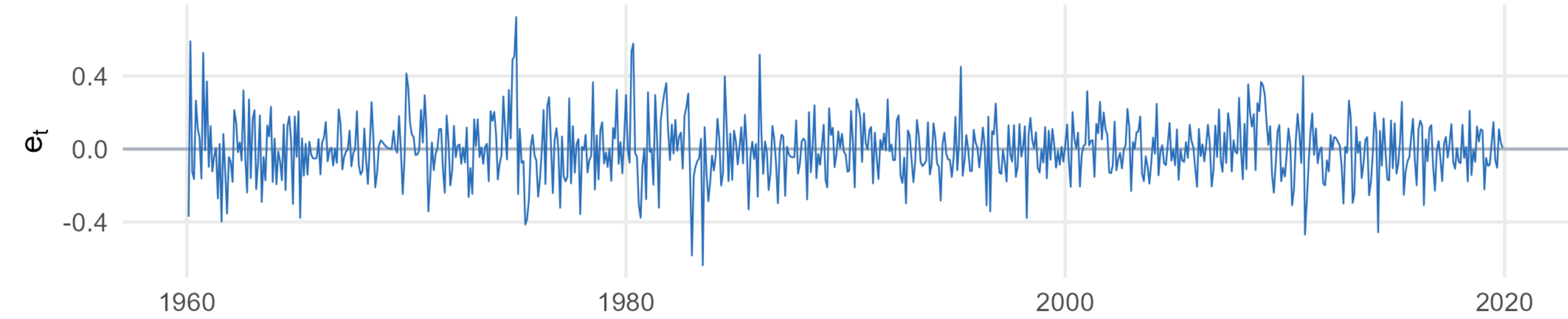
The residual ACF shows no decay from lag 1, no cutoff, no spike at lag 1 — the Module 3 table's first four rows are empty.

What it shows is spikes at 12, 24, 36. Every twelve months. Same sign each time.

Spikes that recur at regular intervals. What are those?

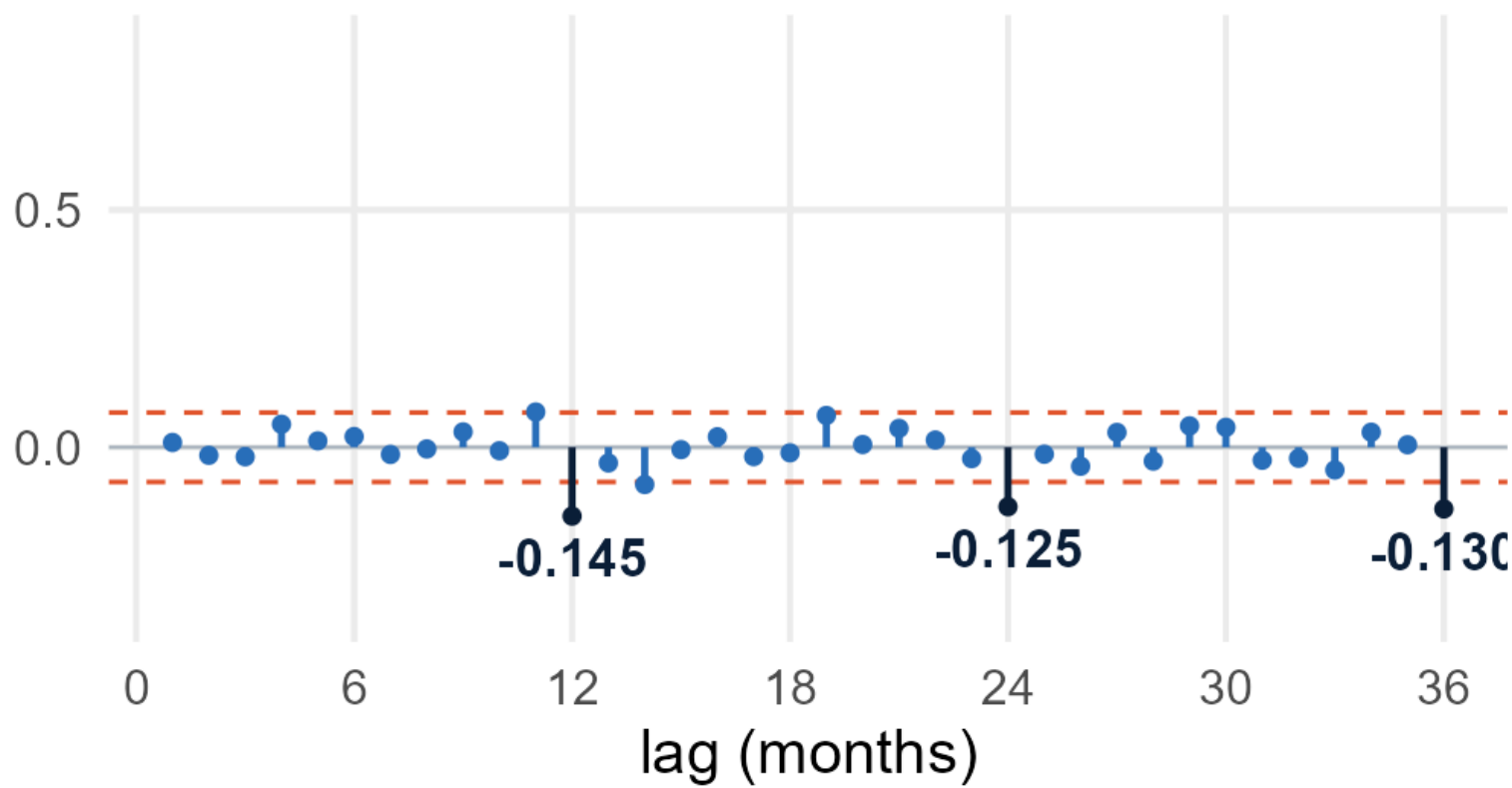
Cached: `checkresiduals()` on the Module 4 fit

Residuals of the Module 4 fit: ARMA(1,2) on `diff(UNRATE)`

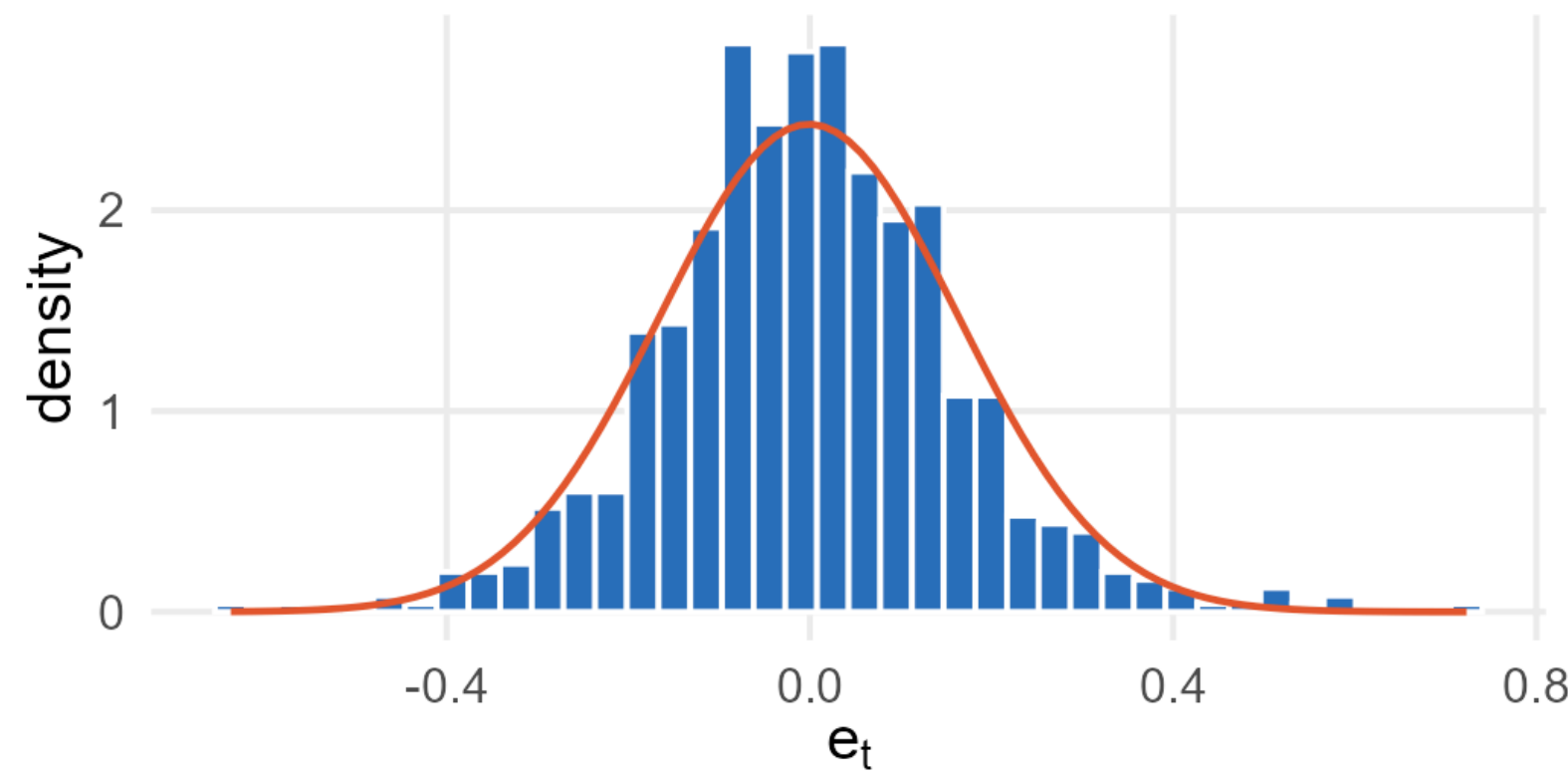


Residual ACF

LB: $h = 10$ $p = 0.807$ pass | $h = 24$ $p = 0.0012$ fail (fitdf 3)



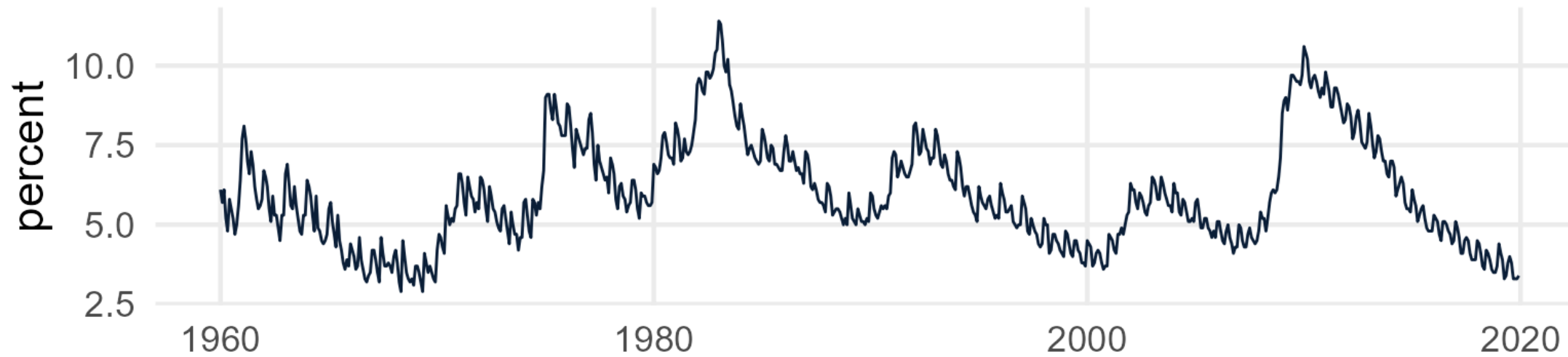
Residual histogram, normal overlay



You can see it

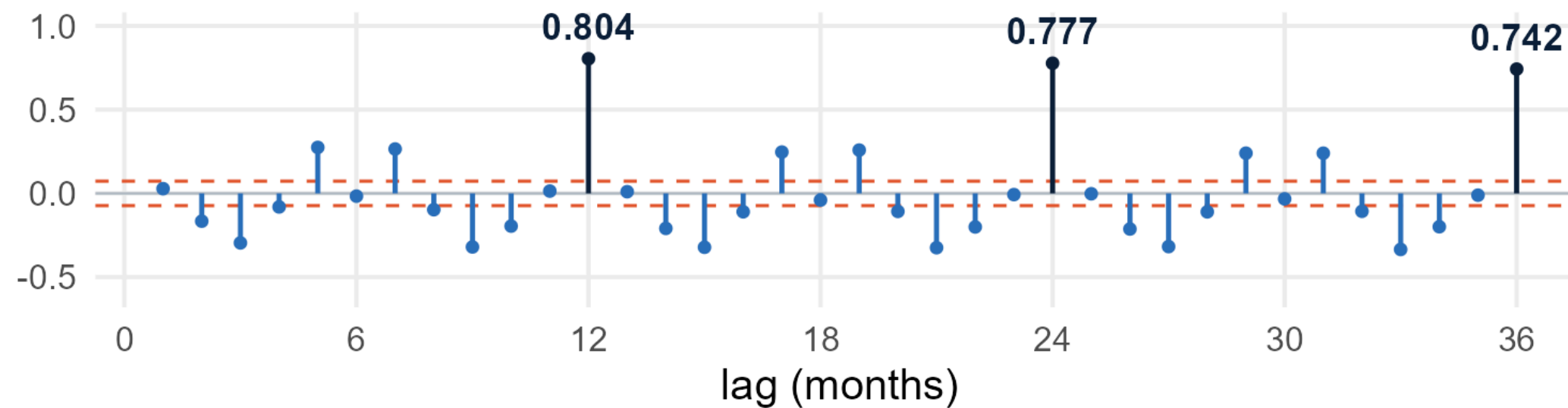
UNRATENSA, not seasonally adjusted, 1960-2019

the raw series -- what the survey measures before any procedure touches it



Sample ACF of diff(UNRATENSA)

no model fitted -- this is the series' own autocorrelation



UNRATENSA — the same civilian unemployment rate, *not* seasonally adjusted. What the survey measures each month.

$\hat{\rho}(12)$ of $\Delta\text{UNRATENSA}$ **0.804**

$\hat{\rho}(24)$ **0.777**

$\hat{\rho}(36)$ **0.742**

band **± 0.073**

$\hat{\rho}(12)$ of ΔUNRATE , published **-0.089**

No model has been fitted. This is the series' own **autocorrelation**. The lag-12 spike in the Module 4 residuals was **serial correlation** — a verdict on a fit. Same picture, different owner. Ten times the band, and barely decaying from one year to the next. Hold that thought.

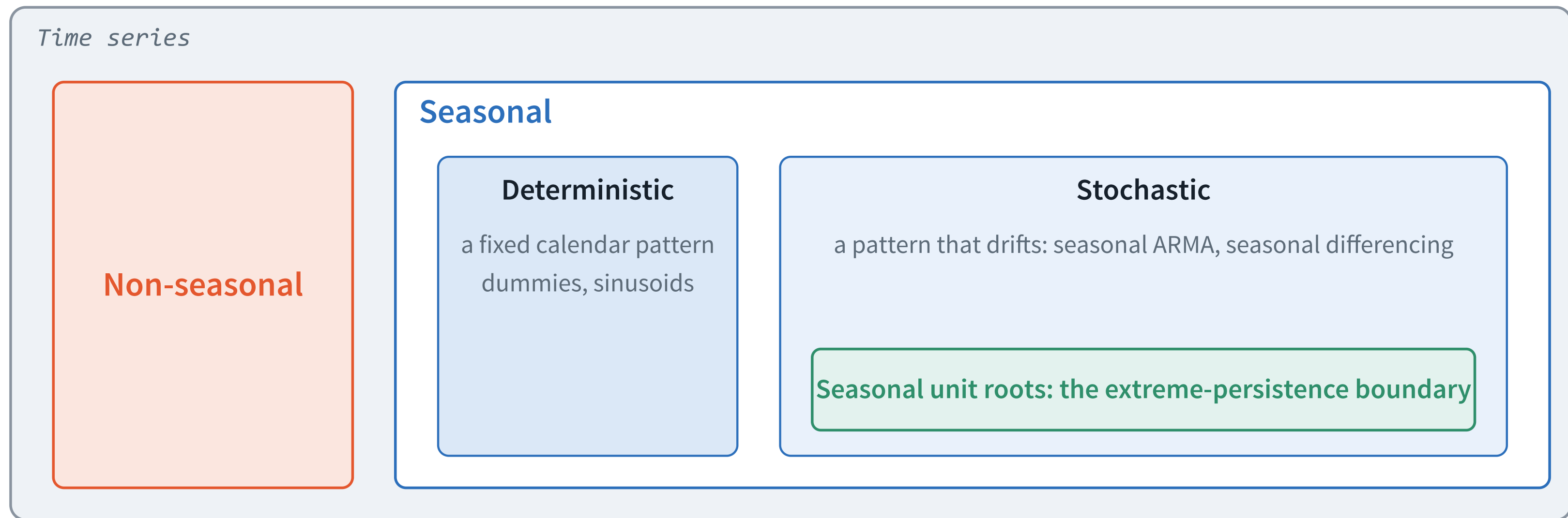
Nobody agrees what it is — but you can see it

Tradition	Core object	Representative work
Calendar means (regularity and repetition)	Seasonal means across a calendar partition	Falkner (1924); Kallek (1978)
Seasonal unit roots and stability (the ARIMA polynomial)	Roots of the AR polynomial at the seasonal frequencies	Hylleberg, Engle, Granger & Yoo (1990); Canova & Hansen (1995)
Unobserved components (signal extraction)	A latent seasonal component S_t identified by a restriction	Hillmer & Tiao (1982); Bell & Hillmer (1984); X-13ARIMA-SEATS
Spectral (frequency domain)	A functional of the spectral density at the seasonal frequencies	Nerlove (1964); Granger (1978)

Each is internally consistent and answers a well-posed question. **None nests the others.** Analysts consult several at once and quietly assemble a superset nobody has written down.

“Some of us looking for answers regarding seasonal analysis may feel as if we’re in a dark room looking for a black cat. That’s bad, but it’s not as bad as it could be. Pity the philosophers who are in a dark room looking for a black cat that isn’t there.” — Zellner et al. (1978, p. 451)

The taxonomy: where the seasonal unit root sits



The seasonal unit root is not a third species. It is the stochastic case pushed to the limit where the seasonal pattern never mean-reverts — the random walk is the AR(1) at $\phi_1 = 1$ (Module 2), and this is the seasonal analogue at $\Phi_1 = 1$. The differencing decision D is the decision about *which side of that boundary* the series is on.

There is no clean “isn’t”

Students want a checklist: seasonality is X; calendar effects, trading days, holidays are *not* seasonality. The course declines, and the reason matters.

- **A definition is a line you draw.** Choose one — seasonal means, seasonal unit roots, a latent component, a spectral peak — and whatever lands inside that line is seasonal *by construction*, whether or not you would have called it that beforehand.
- **Trading-day variation** is the standard case: the number of Mondays in a month varies with a rhythm close to, but not at, the annual frequency. Under a spectral definition it leaks into the seasonal frequencies; under a calendar-means definition it lands in the monthly means. The definition decides, and different definitions decide differently. Moving holidays likewise.
- These effects are **not statistically separable** from seasonality without an extra identifying assumption — which is exactly what X-13 supplies when it pre-regresses them out.
- **The one safe non-example:** monthly sampling alone does not make a series seasonal. A random walk observed every month has a seasonal *frequency* at which structure could live. It does not put structure there.

Hold on to “the line you draw”. The Deeper Dive at the end of today draws one explicitly — a partition of the frequency axis — and then asks a testable question relative to it.

The triage: three paths

Diagnosis or need	Path	What it commits you to
The pattern is deterministic — fixed, exogenous, calendar-driven	Path 1 · seasonal dummies (or Fourier terms) in a regression	a fixed shape; if the pattern drifts, the residuals will say so
The pattern is stochastic — it drifts, and you want to <i>model</i> it and forecast with it	Path 2 · SARIMA — seasonal ARMA terms, with seasonal differencing if there is a seasonal unit root	a joint model of both bands of the console
“I need it gone, not modeled ” — a series you can read, compare month to month, or feed to a non-seasonal model	Path 3 · seasonal adjustment — decompose, remove the seasonal, publish the rest	trusting a procedure that is itself a model

The three are not rivals. Agencies run path 3 to publish the series you read in the news; forecasters run path 2 on the raw or the published series; path 1 is the cheap first question — *is the pattern fixed?*

A journalist comparing this month to last month – which path? A forecaster building next year’s fan chart?

All three, next, on one series: the raw unemployment rate.

Path 1: seasonal dummies

On the **first difference** — the course has treated the unemployment rate as $I(1)$ since Module 2 — regress on eleven month indicators plus an intercept, with month index m and January ($m = 1$) as the reference:

$$\Delta y_t = \alpha + \sum_{m=2}^{12} \beta_m \mathbb{1}\{\text{month}_t = m\} + \epsilon_t$$

- **What it assumes:** a *fixed* monthly pattern. The same January jump in 1965 and in 2019. Eleven parameters $\beta_2, \dots, \beta_{12}$, any shape. (The indicator is written $\mathbb{1}\{\cdot\}$ on purpose: D is today's seasonal differencing order, and j is the AR lag index.)
- **What the residuals will tell you:** if the pattern really is fixed, the residual ACF is clean at 12, 24, 36. If it drifts, the seasonal spikes survive — and that verdict sends you to path 2.
- **The parsimonious cousin, one line:** replace the eleven dummies with a few sine–cosine pairs from `forecast::fourier()`. Fewer parameters, same fixedness — and fixedness, not parsimony, is what is about to be tested.

```
1 seasonal_dummies <- function(y, ref = 1) { ... }      # helpers/seasonality.R: the (s - 1) indicator columns
```

Path 1, live: fit the dummies, read the residuals

```

1 nsa <- read.csv("../..data/UNRATENSA.csv")
2 nsa <- subset(nsa, observation_date >= "1960-01-01" &
3               observation_date <= "2019-12-01")
4 unratensa <- ts(nsa$UNRATENSA, start = c(1960, 1),
5                 frequency = 12)
6 d_unratensa <- diff(unratensa)
7
8 source("../..helpers/seasonality.R") # seasonal_dummies()
9 X_month <- seasonal_dummies(d_unratensa) # Jan = reference
10 fit_dum <- lm(d_unratensa ~ X_month)
11 summary(fit_dum)$r.squared             # 0.72
12 e_dum <- ts(residuals(fit_dum), frequency = 12,
13             start = start(d_unratensa))
14 ggAcf(e_dum, lag.max = 36)
15 Box.test(e_dum, lag = 24, type = "Ljung-Box", fitdf = 0)

```

Before the plot: the dummies absorb $R^2 = 0.72$. Will lag 12 clear the band?

residual $\widehat{\rho}_e$ (12)	0.350
residual $\widehat{\rho}_e$ (24)	0.298
residual $\widehat{\rho}_e$ (36)	0.223
band	± 0.073

Honestly: the fitdf = 0 Ljung-Box also rejects for non-seasonal reasons — lags 2, 3, 5, 7, 11 breach too, the Module 4 dynamics a dummy regression cannot carry. Give it ARMA(1,2) errors and only the seasonal lags do the rejecting: $h = 10$ passes ($p = 0.30$), $h = 24$ fails.

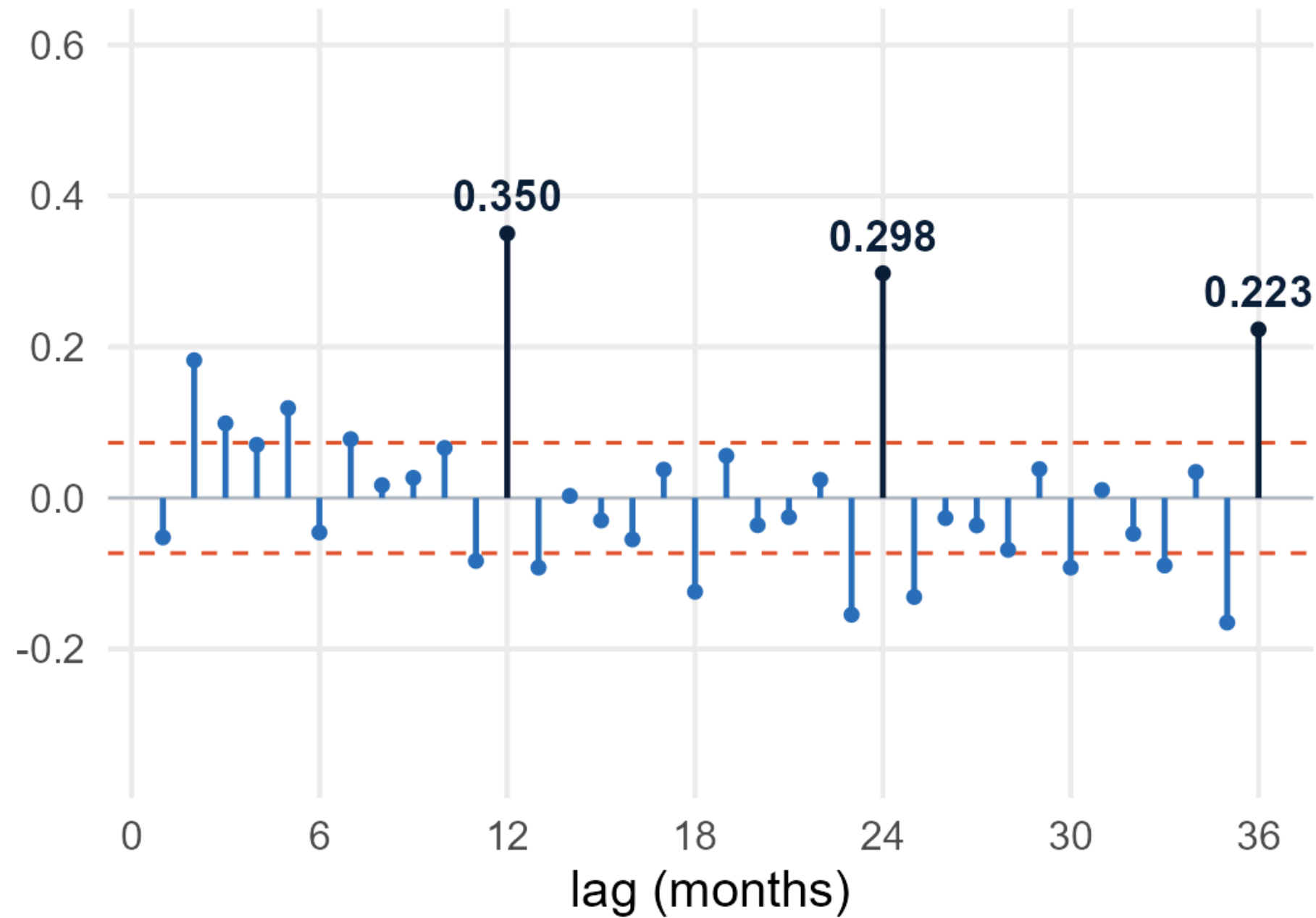
Is the pattern fixed? Month effects by half, 1960–89 against 1990–2019: $F(11, 695) = 12.5$, $p \approx 10^{-21}$. June halves ($+0.84 \rightarrow +0.43$); July flips sign ($-0.33 \rightarrow +0.02$).

The fixed part is absorbed. A drifting part remains. On to path 2.

Cached: the dummy residuals

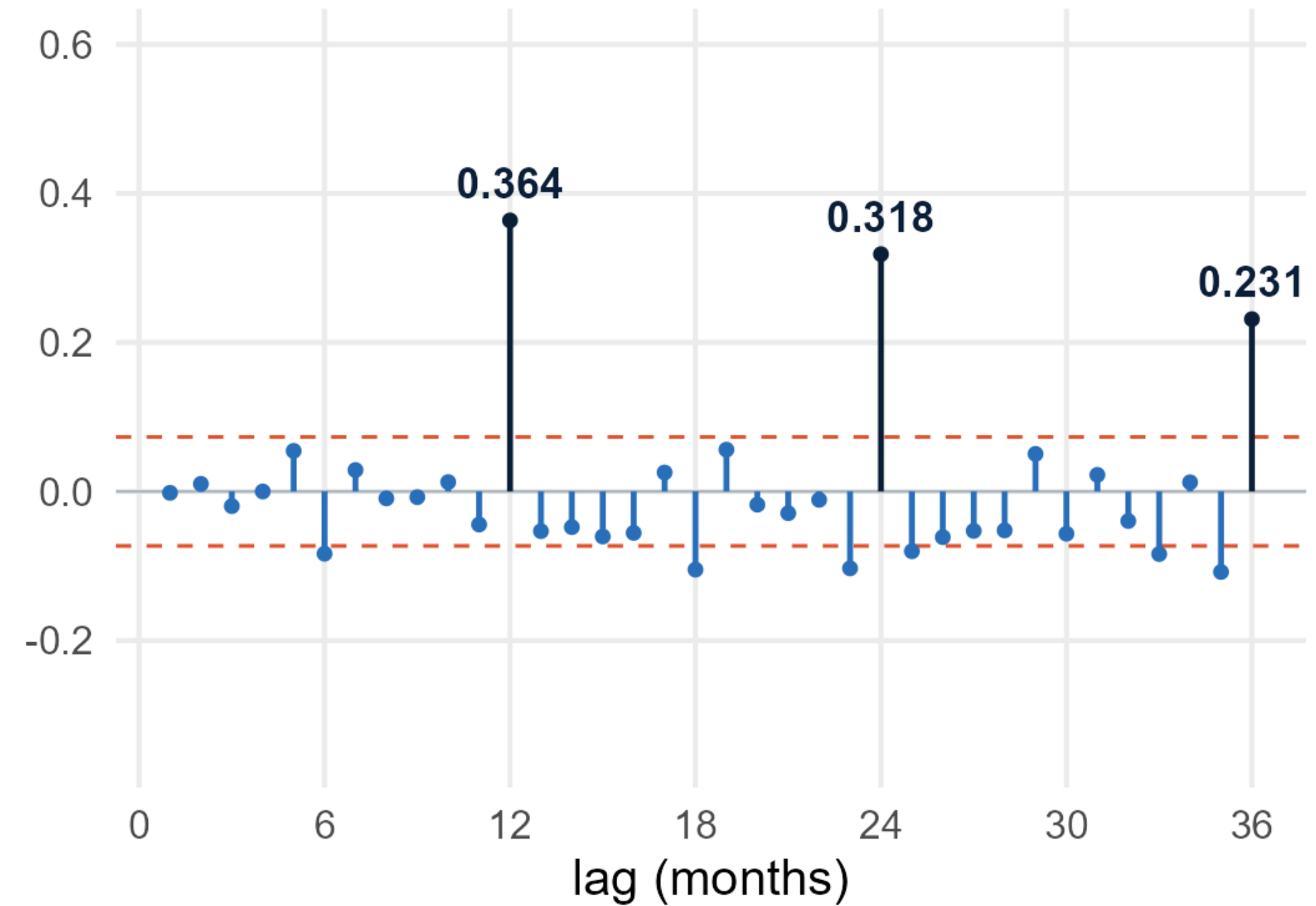
Month dummies + intercept: residual ACF

seasonal lags survive; low lags 2, 3, 5, 7, 11 breach too



Same dummies with ARMA(1,2) errors

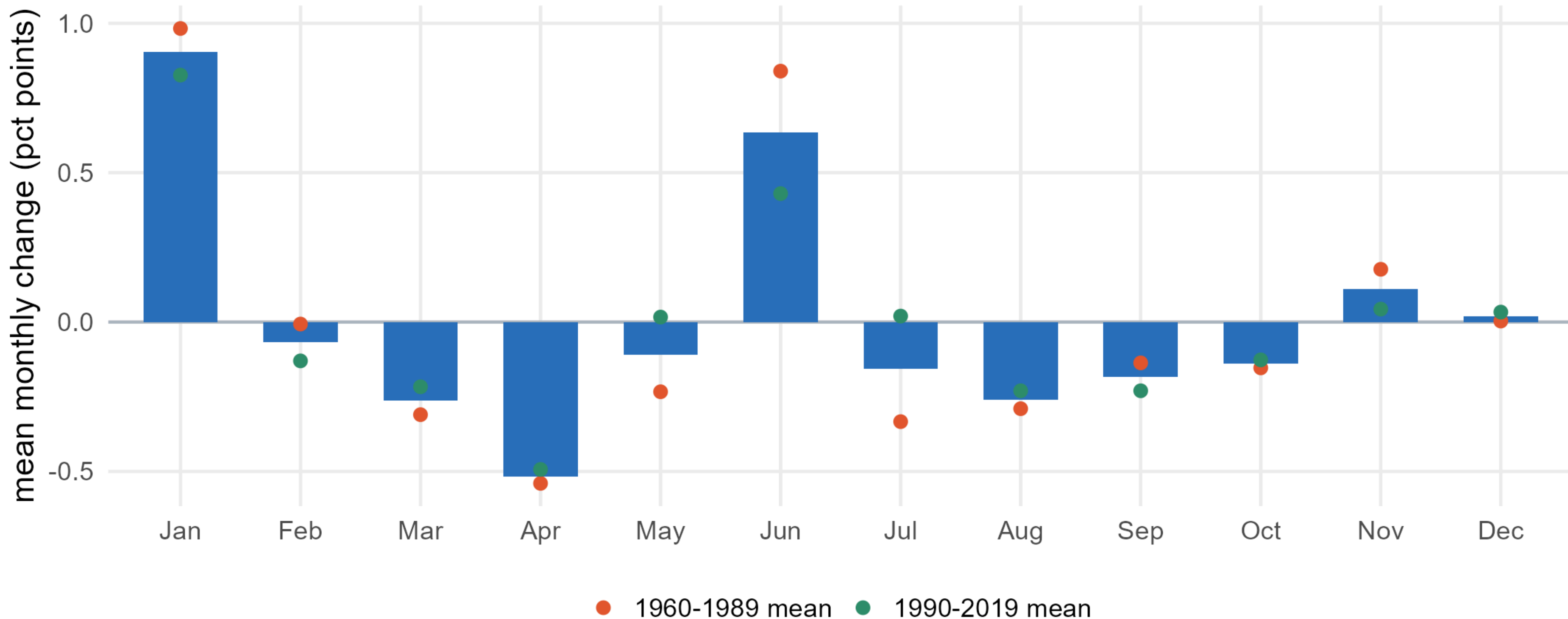
LB (fitdf 3): $h = 10$ $p = 0.30$ pass | $h = 24$ $p = 0$ fail



Cached: the fitted monthly pattern, by half of the window

Path 1: the fitted monthly pattern in diff(UNRATENSA) (R-squared = 0.72)

bars = full-window month means; points = each half. Equal month effects? $F(11, 695) = 12.5$, $p = 1.15e-21$



Path 2: the seasonal band on the console

The non-seasonal band works at lags 1, 2, 3, ... — the knobs since Module 3. A seasonal process has a **second band** at lags $s, 2s, 3s, \dots$ with the same kinds of knobs: AR, MA, and differencing.

$$\Phi_p(L) \Phi_P(L^s) (1-L)^d (1-L^s)^D y_t = \Theta_q(L) \Theta_Q(L^s) \epsilon_t$$

$$\Phi_p(L) = 1 - \phi_1 L - \dots - \phi_p L^p$$

$$\Phi_P(L^s) = 1 - \Phi_1 L^s - \dots - \Phi_P L^{Ps}$$

$$\Theta_q(L) = 1 + \theta_1 L + \dots + \theta_q L^q$$

$$\Theta_Q(L^s) = 1 + \Theta_1 L^s + \dots + \Theta_Q L^{Qs}$$

- **Lowercase** (p, d, q) , ϕ, θ : the base band. **Uppercase** (P, D, Q) , Φ, Θ : the seasonal band. Minus signs on AR, plus signs on MA, in both bands.
- s is the **period**: 12 monthly, 4 quarterly, 7 daily-with-a-weekly-cycle. $\text{ARIMA}(1, 1, 1)(0, 1, 1)_{12}$ reads: base band, seasonal band, monthly.
- $(1 - L^s)^D$ is **seasonal differencing**, $\Delta_s y_t = y_t - y_{t-s}$ — the Module 2 operator at the seasonal frequency. New today.
- When the seasonal MA order appears alone in prose it is written Q_s , so it does not collide with the Ljung-Box Q .

Multiplicative structure: lag 13 for free

The bands **multiply**. Take the smallest case with both on, $\text{ARIMA}(1, 0, 0)(1, 0, 0)_{12}$:

$$\begin{aligned} & \left(1 - \phi_1 L\right) \left(1 - \Phi_1 L^{12}\right) y_t = \epsilon_t \\ & 1 - \phi_1 L - \Phi_1 L^{12} + \phi_1 \Phi_1 L^{13} \end{aligned}$$

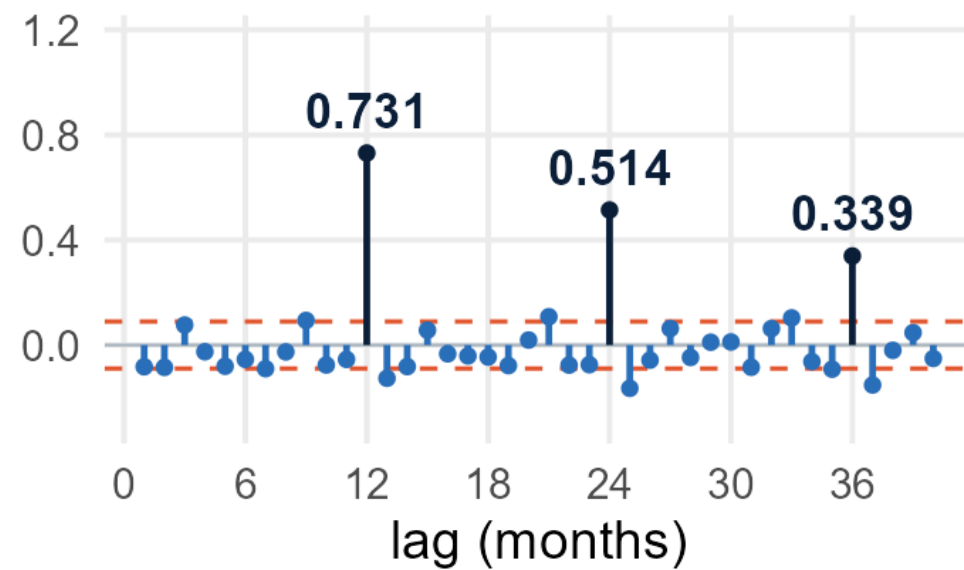
- Coefficients at lags 1, 12, and **13**. The lag-13 term is not a free parameter — it is the product $\phi_1 \Phi_1$, forced by the multiplication.
- Lag-12 *and* lag-13 dynamics for **two parameters** instead of thirteen. That is what makes SARIMA parsimonious next to a long AR that reaches lag 12 by sheer depth.
- The cross-frequency dynamics come for free as products of low-order terms. If the residuals disagree with the product, a free lag-13 term (an additive seasonal) is the less restrictive alternative — but for most macro applications the product works.

`sarima_simulator()` in `helpers/simulators.R` does exactly this expansion before running the master-equation loop. (Worked in the notes' Deeper Dive §5.7.2.)

Reading the seasonal ACF: decaying or flat?

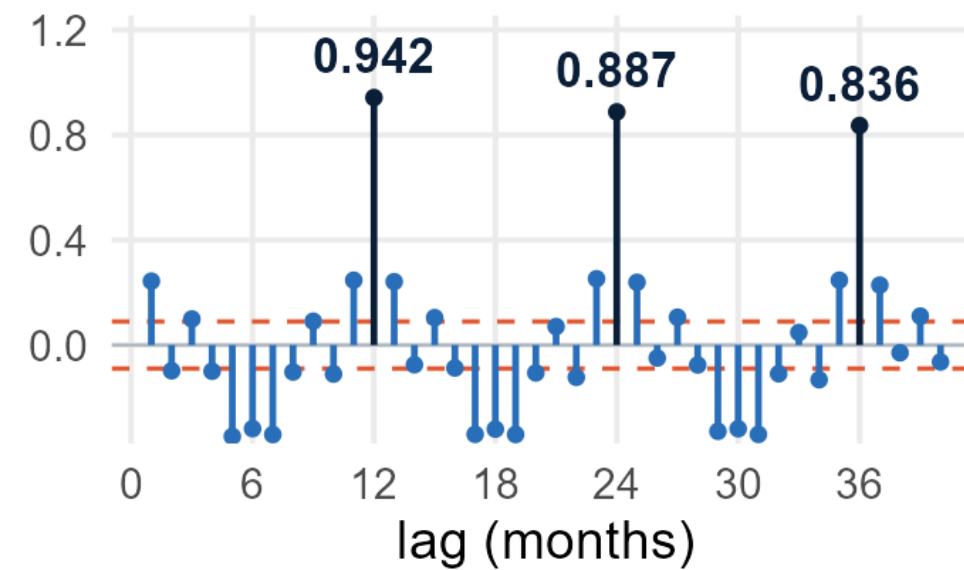
Decaying (Phi = 0.8)

seasonal AR(1): spikes shrink $\rightarrow D = 0$



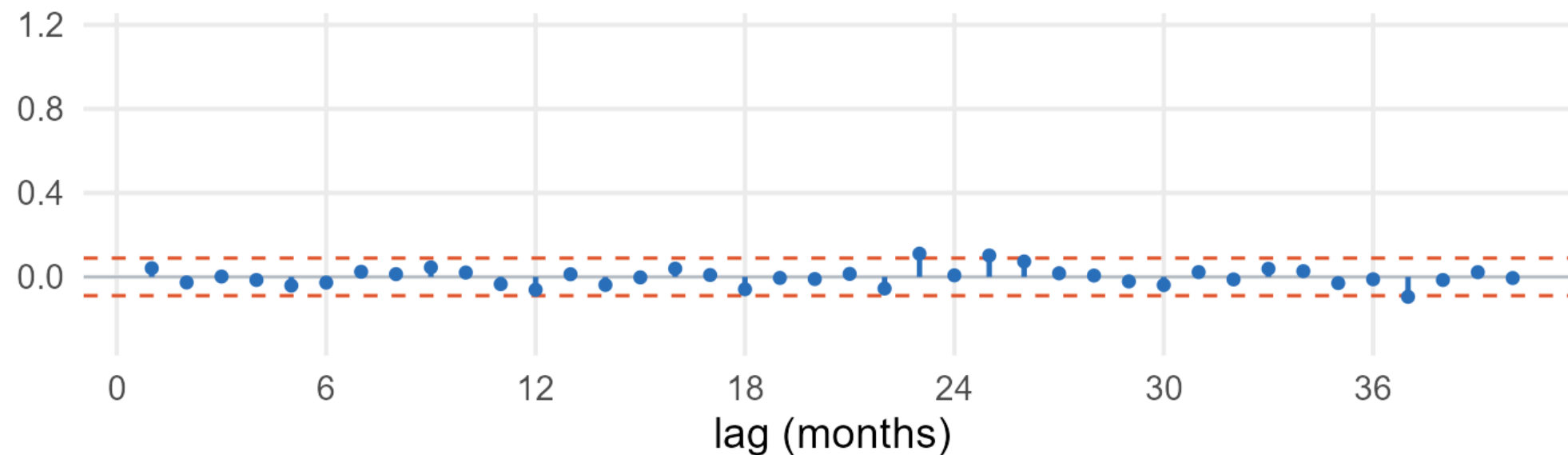
Flat (Phi = 1)

seasonal random walk: spikes stay $\rightarrow D = 1$



The flat case after one seasonal difference

$\Delta_{12} y_t = y_t - y_{t-12}$: the seasonal spikes are gone



Decaying spikes at $s, 2s, 3s$: a stationary seasonal ARMA. Model the band, keep $D = 0$.

Flat spikes near one: a seasonal unit root. Difference once at the seasonal frequency, $D = 1$, then re-read.

Do not reach for $D = 2$ without fresh evidence of a second seasonal unit root.

The benchmark, Box & Jenkins's airline model, ARIMA $(0, 1, 1) (0, 1, 1)_{12}$:

$$(1 - L) (1 - L^{12}) y_t = (1 + \theta_1 L) (1 + \theta_{12} L^{12}) \epsilon_t$$

Two differences, two parameters. A benchmark to improve on, not an answer.

Path 2, live: four candidates, one decision

```

1 fit_air  <- Arima(unratensa, order = c(0, 1, 1), seasonal = list(order = c(0, 1, 1), period = 12))
2 fit_d0   <- Arima(unratensa, order = c(1, 1, 2), seasonal = list(order = c(1, 0, 1), period = 12))
3 fit_auto <- auto.arima(unratensa)
4 fit_win  <- Arima(unratensa, order = c(1, 1, 2), seasonal = list(order = c(0, 1, 1), period = 12))
5 source("../helpers/diagnostics.R") # lb_both()
6 lb_both(fit_win); checkresiduals(fit_win)

```

Before the table: which of the four will put a coefficient on a boundary?

Candidate	Seasonal coefficient	LB p (10 / 24)	Verdict
Airline $(0, 1, 1) (0, 1, 1)_{12}$	$\widehat{\theta}_1 = -0.78$, interior	$10^{-13} / 10^{-11}$	fails both: the $(0, 1, 1)$ block is too thin — residual ACF 0.21, 0.14, 0.12, 0.11 at lags 2–5
$D = 0$ $(1, 1, 2) (1, 0, 1)_{12}$	$\widehat{\phi}_1 = 0.994$	0.97 / 0.76	passes and wins the $D = 0$ table, but the seasonal AR root, in $z = L^{12}$, has modulus 1.006, within 0.01 of the unit circle: a seasonal unit root in disguise
<code>auto.arima()</code> $(2, 0, 2) (2, 1, 2)_{12}$	$d = 0, D = 1$, eight coefficients	0.27 / 0.47	passes; a third differencing class, so its AICc is on neither table — a very good intern, read its work
Winner $(1, 1, 2) (0, 1, 1)_{12}$	$\widehat{\theta}_1 = -0.75$, interior	0.88 / 0.42	passes both horizons, every coefficient interior, same differencing as the airline

Information criteria compare only within a differencing class

A $D = 1$ likelihood is evaluated on the seasonally differenced series — a different, shorter series than a $D = 0$ model uses — and $d = 0$ is different again. The four candidates fall into **three classes**, and rows are comparable only within a class.

The $(d, D) = (1, 1)$ class, the airline model against the winner:

	AIC	BIC	LB p (10 / 24)
airline $(0, 1, 1) (0, 1, 1)_{12}$	-155.7	-142.0	$10^{-13} / 10^{-11}$
$(1, 1, 2) (0, 1, 1)_{12}$	-206.2	-183.4	0.88 / 0.42

The winner beats the airline by **50 AIC / 41 BIC** points — the whole gap is the non-seasonal block.

- In the **$(1, 0)$ class** the seasonal ARMA(1,1) band wins by hundreds of points over Module 4's bare block — and its winner is the model with $\hat{\Phi}_1 = 0.994$.
- `auto.arima()` sits alone in the **$(0, 1)$ class**; its AICc is on neither table.

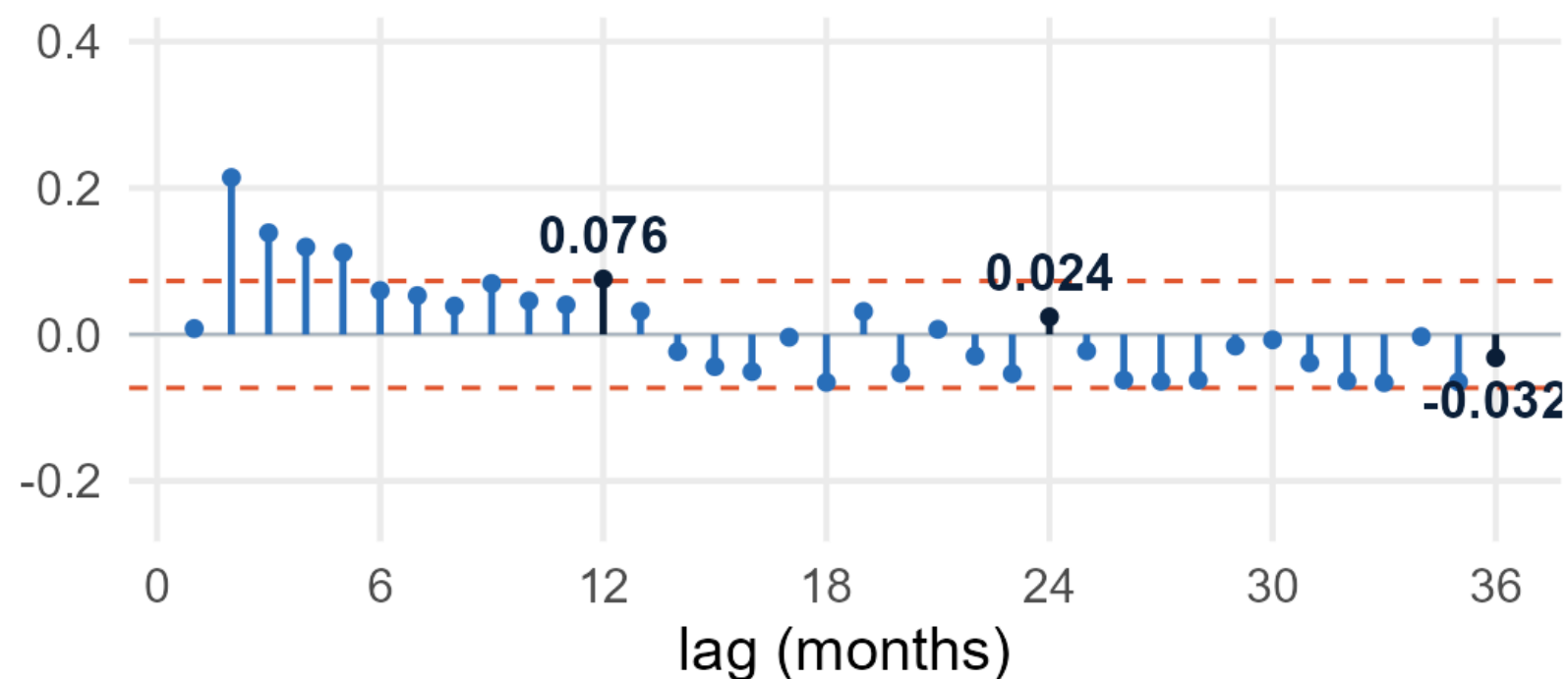
The tables cannot pick among candidates 2, 3, 4. Diagnostics and the coefficients do.

And a check nobody planned: X-13's own automatic model selection, run on the next path with no guidance, chose **the same orders** as the winner.

Cached: the four residual ACFs

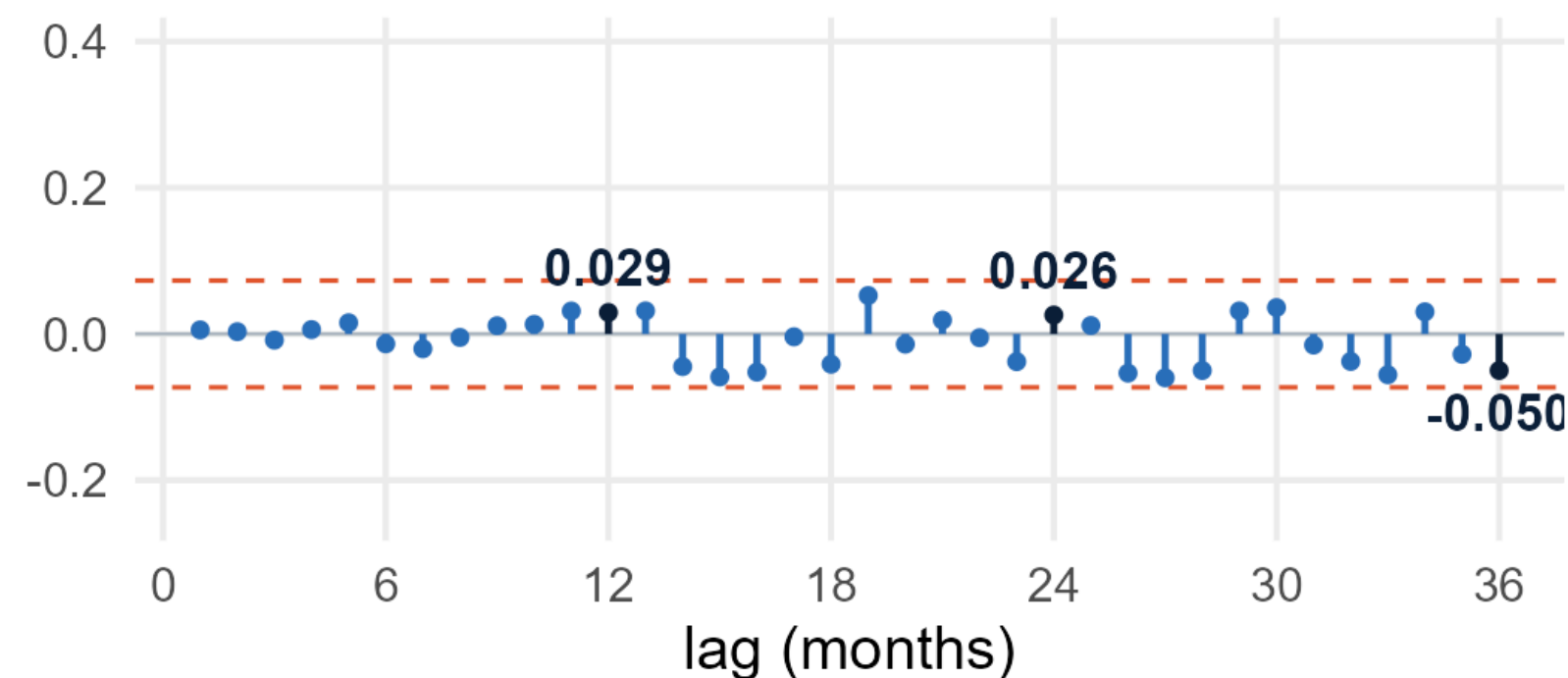
Airline (0,1,1)(0,1,1)[12]: low lags fail

Theta1 = -0.78 interior; LB p = 1.54e-13 / 2.48e-11



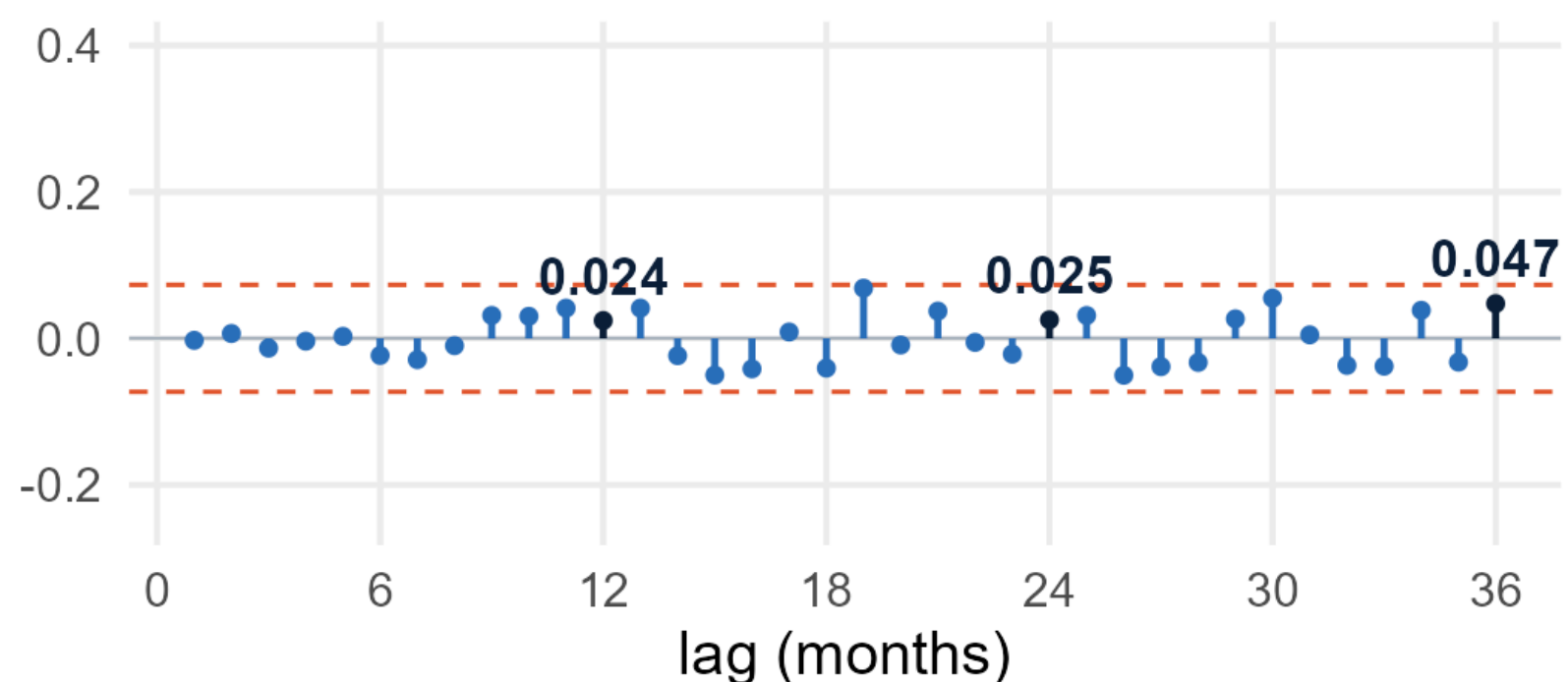
D = 0: (1,1,2)(1,0,1)[12], Phi1 = 0.994

LB p = 0.97 / 0.76; seasonal AR root modulus 1.006



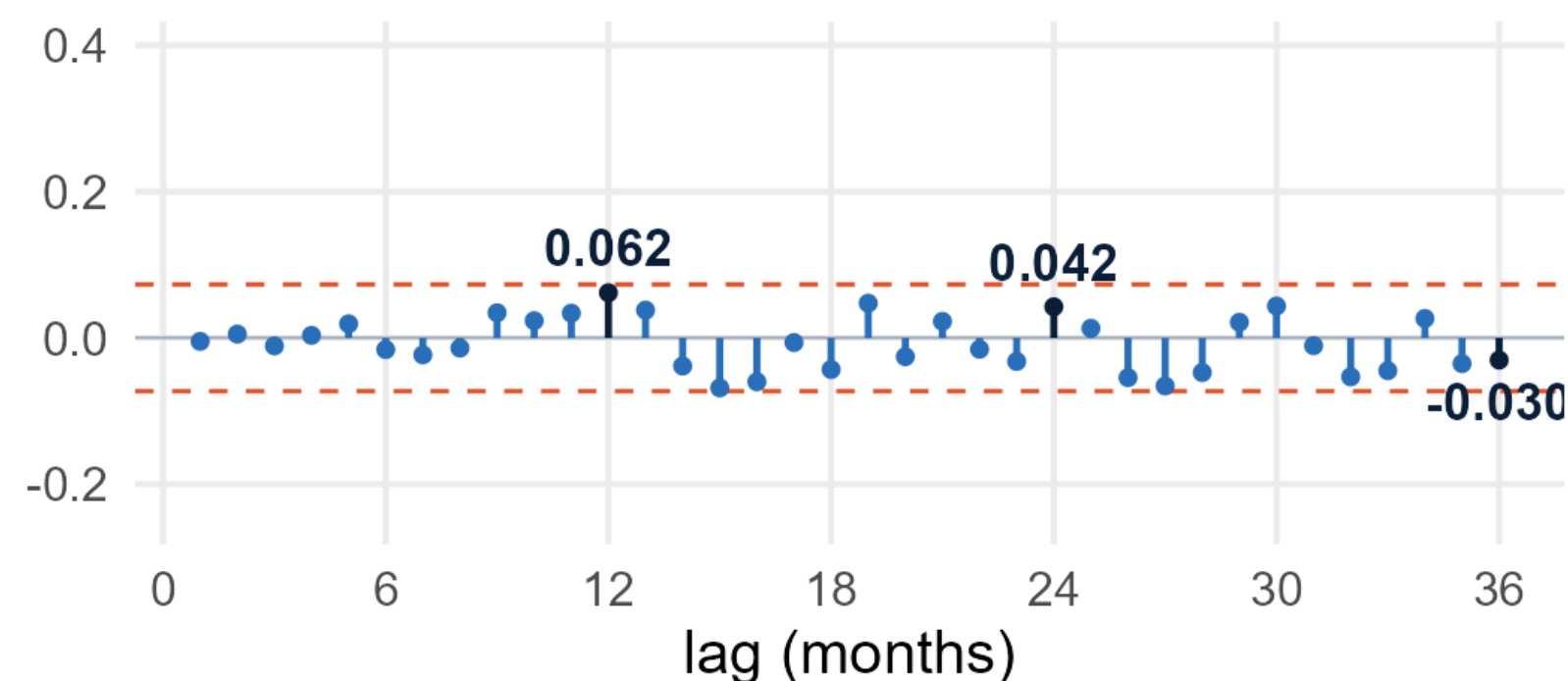
auto.arima(): (2,0,2)(2,1,2)[12], d = 0, D = 1

8 coefficients; LB p = 0.27 / 0.47



Winner (1,1,2)(0,1,1)[12], same D as airline

Theta1 = -0.75 interior; LB p = 0.88 / 0.42



D is a commitment, and the data push back from both sides

One seasonal difference too many

On the **published, already adjusted** UNRATE, impose $D = 1$ — the airline model:

-0.99999

$\hat{\theta}_1$, AIRLINE ON UNRATE

The seasonal MA coefficient is pinned at the non-invertibility boundary: the Module 3 over-differencing signature. The model is trying to undo a difference the data did not need — the adjustment procedure had, in effect, already taken it.

One seasonal difference too few

On the **raw** UNRATENSA, leave $D = 0$ and model the band with seasonal ARMA terms:

+0.994

$\hat{\phi}_1, (1, 1, 2) (1, 0, 1)_{12}$ ON UNRATENSA

The seasonal AR coefficient is pinned at one — the root in $z = L^{12}$ has modulus 1.006, within 0.01 of the unit circle. The model is trying to manufacture a difference the data *did* need. Diagnostics pass; the coefficient tells the truth.

Either boundary estimate says: revisit D , not the ARMA orders.

Neither shows up in an IC table, because IC cannot compare across differencing orders — a $D = 1$ likelihood is evaluated on a different, shorter series. When candidates disagree about differencing, decide on diagnostics and coefficients, and on out-of-sample forecasts (Module 6).

Path 3: what seasonal adjustment does

A forecaster wants to *model* the seasonal band. A statistical agency, a journalist, a policymaker wants it *gone*, so this month can be compared with last month without the January jump in the way.

$$y_t = \text{trend-cycle}_t + \text{seasonal}_t + \text{irregular}_t$$

- **What it does.** Treat the series as three latent pieces (the unobserved-components tradition). Estimate the seasonal piece with a model. Subtract it. Publish the rest: trend-cycle plus irregular. The seasonal piece is *what the procedure decided was seasonal* — under its definition, with its identifying restriction.
- **Who does it.** The BLS, the Census Bureau, the BEA and their counterparts abroad run **X-13ARIMA-SEATS** (Census Bureau). It fits a regARIMA model — a SARIMA plus regressors for trading days, leap years, holidays and outliers — extends the series at both ends, and extracts the seasonal by X-11 filters or a SEATS model-based decomposition. Every option has a default; agencies tune them per series.
- **Why you must recognize it.** Most headline macro series you will ever download — the unemployment rate, payrolls, retail sales, GDP, CPI — are its outputs.

Recognition level, deliberately: no simulate step, no spec file, one call.

Path 3, live: one call, and the reveal

```

1 library(seasonal)           # wraps X-13ARIMA-SEATS
2 fit_x13 <- seas(unratensa)   # no options, no spec file
3 summary(fit_x13)            # chose (1 1 2)(0 1 1), SEATS,
4                             # no log, leap year + weekday,
5                             # and a level shift at 1975-01
6 sa_x13 <- final(fit_x13)
7 autoplot(cbind(published = unrate, x13 = sa_x13))
8 cor(sa_x13, unrate)
9 mean(abs(sa_x13 - unrate)); max(abs(sa_x13 - unrate))

```

Before the overlay: how close will one default call get to the published line?

correlation with published UNRATE	0.9985
-----------------------------------	---------------

mean difference	0.065 pp
------------------	-----------------

max difference (1976-02)	0.37 pp
----------------------------	----------------

months within 0.1 pp	80%
----------------------	------------

The two adjusted lines are one line. The gaps are what a default call against a production run should leave — BLS adjusts concurrently, with its own span, regressors and revision history. One X-13 call **reproduces** the published series. Not *equals*.

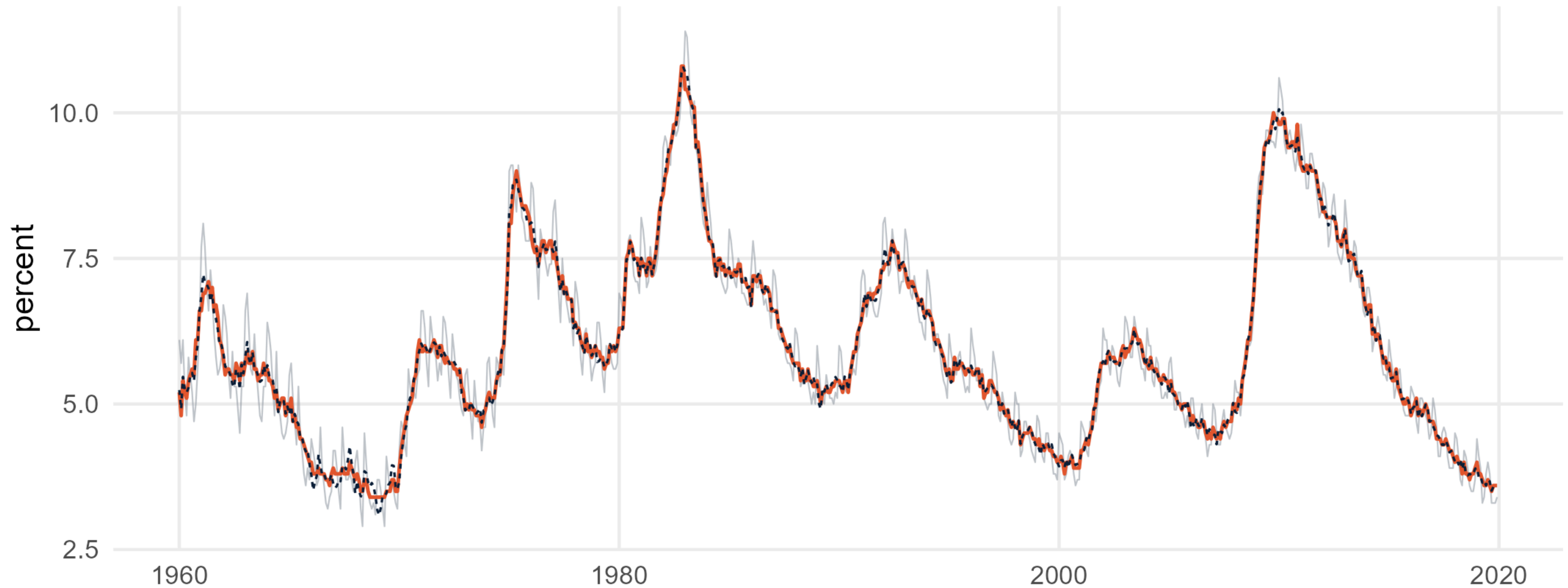
The UNRATE series you have used since Module 2 is the output of this procedure.

Every ACF, ADF test, ARMA fit and information criterion in Modules 2–4 was computed on trend-cycle-plus-irregular from an X-13 run at the BLS, not on what the survey measured. Not a complaint — it changes what you are modeling.

Cached: the overlay

One X-13 call on the raw series, against the published UNRATE

correlation 0.9985 | mean |difference| 0.065 pp | max |difference| 0.37 pp (1976-02)



--- seas(unratensa): one default X-13 call — UNRATE (published, seasonally adjusted) — UNRATENSA (raw)

The cost

Adjustment is a model. It fits a SARIMA, imposes an identifying restriction, and removes what that restriction calls seasonal. Like any model it can be wrong relative to the data — so the rule from Part A applies to it exactly as it applies to your own fits.

- The adjustment procedure can have **serial correlation of its own**. When it shows up at the seasonal lags of the published series, the course's name for it is **residual seasonality** — seasonal autocorrelation left in a series that has already been adjusted.
- The published series carries the procedure's choices as well as the data's dynamics. A model fitted to it is partly modeling the procedure.
- The field's tests and its filters answer different questions. A diagnostic from one tradition can flag seasonality that a filter built in another was never designed to remove — and the two do not speak the same language.

You cannot diagnose residual seasonality without saying what seasonality *is*.

A test for “seasonality left over” needs a definition to test against — the one thing the field has never agreed on. The Deeper Dive that follows is one way of drawing that line so it can be tested. It is optional and never assessed.

Two representations of one series

Deeper Dive · not assessable

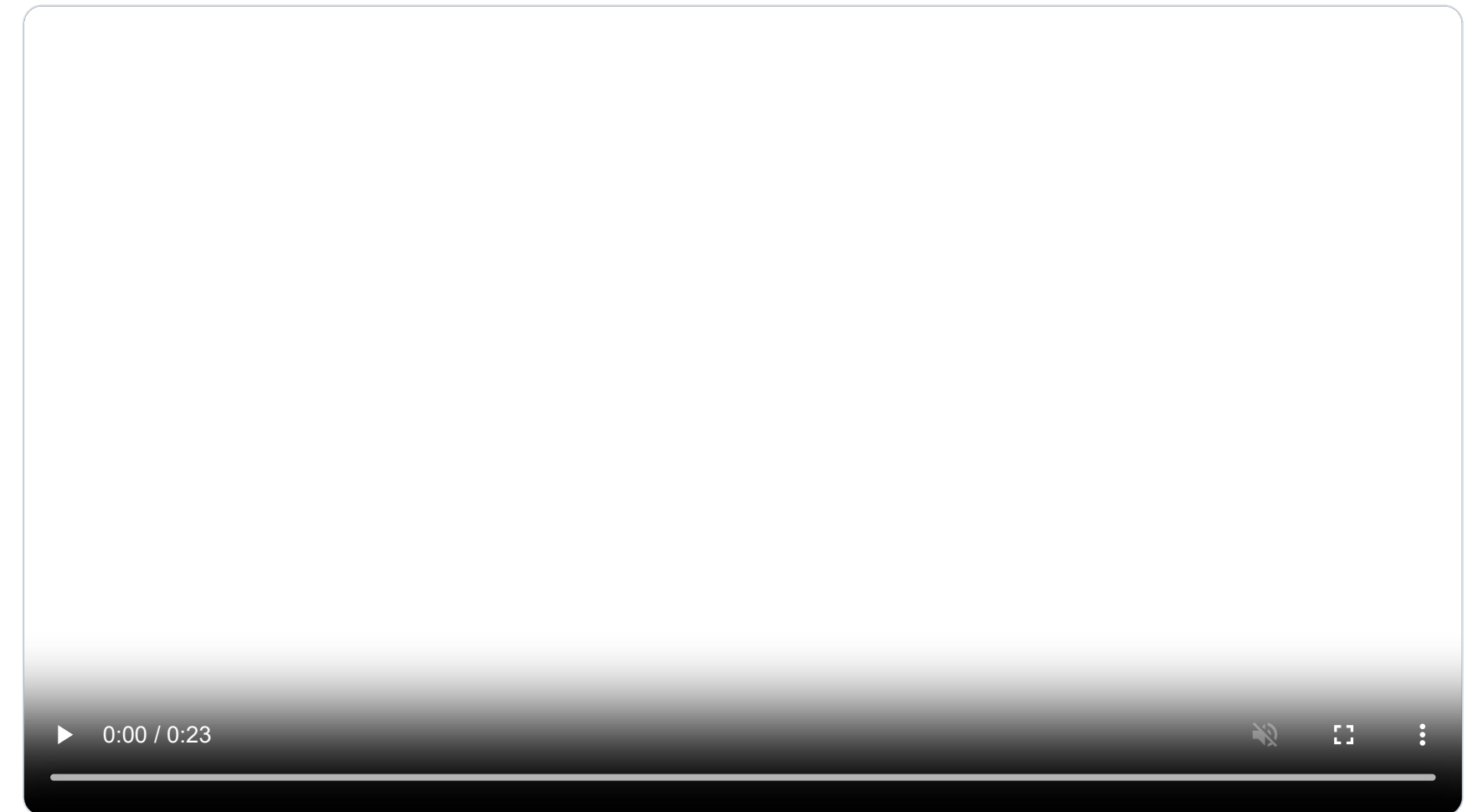
Every finite series $\{x_t\}_{t=1}^T$ has two equivalent descriptions, connected by the **discrete Fourier transform** at the Fourier frequencies

$\omega_j = 2\pi j/T$:

$$d_x(\omega_j) = \sum_{t=1}^T x_t e^{-i\omega_j t} \quad x_t = \frac{1}{T} \sum_{j=0}^{T-1} d_x(\omega_j) e^{i\omega_j t}$$

- The **time domain** orders observations by calendar position — the language of every ACF and lag polynomial in this course.
- The **frequency domain** reallocates the *same* variance by **rate of repetition**: every two months, every six, every year, every decade.
- **Lossless**. The second formula recovers the series exactly.

The series is a mixture. The DFT is the centrifuge that separates it into pure tones. For monthly data the annual cycle lives at $\omega = \pi/6$, with **harmonics** at $\pi/3, \pi/2, 2\pi/3, 5\pi/6, \pi$.



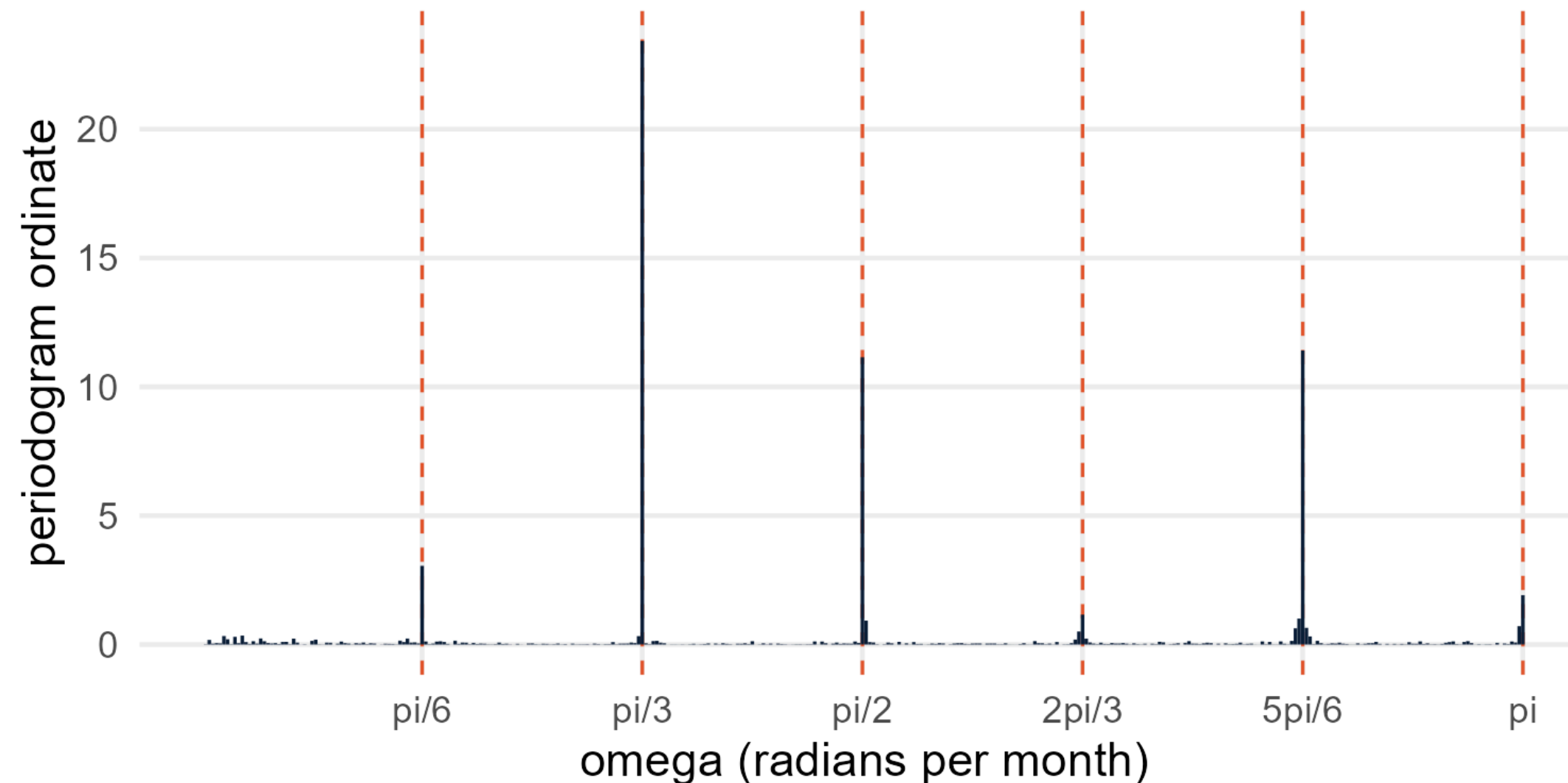
Time to frequency: the instructor's animation, local file. Press play.

The same seasonality in both domains

Deeper Dive · not assessable

Periodogram of diff(UNRATENSA)

dashed = six seasonal harmonics; tallest ordinate at $\pi/3$, not $\pi/6$



The **periodogram** $I_T(\omega_j) = \frac{1}{T}|d_x(\omega_j)|^2$ is the sample estimate of how variance is spread across frequencies.

Here it is for $\Delta\text{UNRATENSA}$ — the series whose ACF spiked at 12, 24, 36. The spikes sit on the dashed lines: the same seasonality, by rate of repetition.

Which line is tallest? $\pi/3$ — the **semiannual** harmonic, a January jump and a June jump — not the annual $\pi/6$. The lag-12 spike cannot tell you that; the periodogram shows it at a glance.

Model-free: deterministic, stochastic and unit-root seasonality all show up the same way.

A definition you can test

Deeper Dive · not assessable

Draw the line *before* looking at the data:

1. **Choose the cycles of interest.** Monthly data, annual period: the six harmonics $\Omega = \{\pi/6, \pi/3, \pi/2, 2\pi/3, 5\pi/6, \pi\}$.
2. **Partition** $(0, \pi]$ into M equal-width bins $B_1, \dots, B_M$. A bin is **seasonal** if it contains a harmonic; call that set $\mathcal{M}_1$, the rest $\mathcal{M}_0$.
3. **Compare the tallest peaks.** With $b_m(f)$ the highest value of the spectral density f inside bin m ,

$$\Delta(f) = \max_{m \in \mathcal{M}_1} b_m(f) - \max_{m \in \mathcal{M}_0} b_m(f)$$

A series is seasonal, relative to the chosen cycles and partition, if and only if $\Delta > 0$.

In plain language: **the chosen cycles are demonstrably larger than every other cycle.** Relative, not absolute, so a narrow spike at $\pi/6$ and a diffuse hump around $\pi/3$ count alike, with no commitment to what generated the peak. Whatever lands in a seasonal bin counts, trading-day leakage included — the “no clean isn’t” made operational.

One line on notation. Δ has meant the difference operator since Module 1, $\Delta y_t = y_t - y_{t-1}$. This Δ is a different object — a difference of two peak heights — and its sample version $\widehat{\Delta}$ is a test statistic. The tell: $\widehat{\Delta}$ never acts on a y_t .

The test: a null distribution in closed form

Deeper Dive · not assessable

Replace f with the periodogram and take the same two maxima:

$$\widehat{\Delta} = \max_{m \in \mathcal{M}_1} b_m(I_T) - \max_{m \in \mathcal{M}_0} b_m(I_T)$$

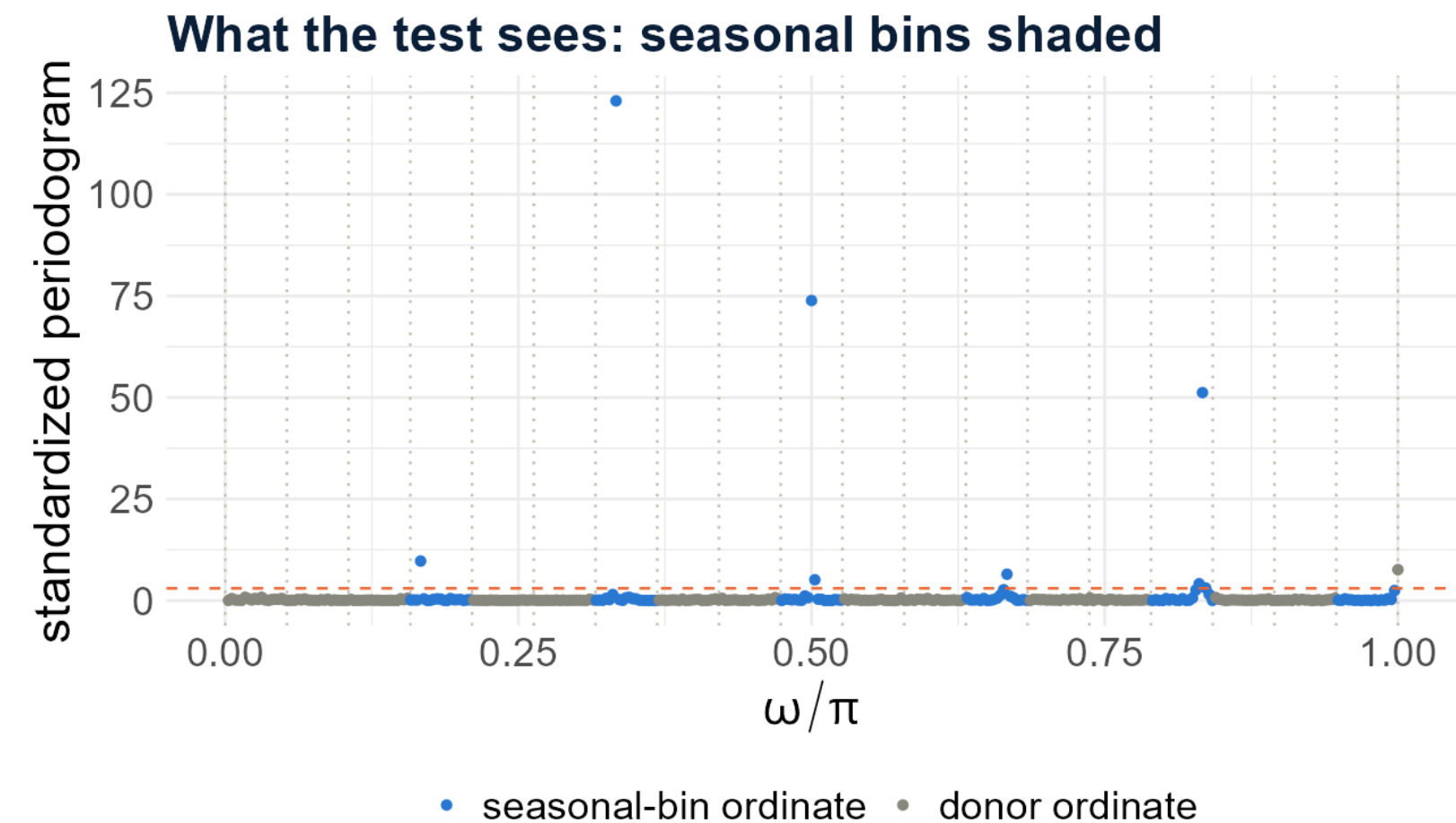
Under white noise with innovation variance σ^2 , periodogram ordinates are approximately independent exponentials; the max of N of them is Gumbel, and a difference of two Gumbels with a common scale is **Logistic**. With N_1 seasonal and N_0 non-seasonal ordinates,

$$\widehat{\Delta} \mid H_0 \Rightarrow \text{Logistic}\left(\sigma^2 \log \frac{N_1}{N_0}, \sigma^2\right)$$

Closed form: no simulation, no bandwidth, no kernel. (*The paper writes τ ; the course writes σ^2 .*)

Prewhiten first with a BIC-chosen non-seasonal ARIMA, so ordinary autocorrelation cannot colour the non-seasonal maximum; **standardize** to unit variance, so $\sigma^2 = 1$.

```
1 remotes::install_github("grycrnwl/freqseas") # once
2 library(freqseas)
3 test_nsa <- seas_test(unratensa)             # defaults
4 summary(test_nsa)
```



$\widehat{\Delta}$ on UNRATENSA	122.2
5% critical value	2.19
M bins · whiteners	19 · $d = 1$, AR(2)
verdict	reject: seasonal

Adjustment as the logical conjugate

Deeper Dive · not assessable

Most shops run a test from one tradition and a filter from another, so the filter removes an object the test never measured. The requirement here: **the adjustment removes exactly the excess the test detects**, so re-running the test on the output fails to reject *by construction*. **Stochastic spectral imputation** (SSI), in three moves:

1. **Setup.** DFT the whitened series. Seasonal-bin ordinates are the **targets**; quiet ordinates below a chosen quantile of the non-seasonal floor are the **donors**.
2. **Impute.** Reset each target's *magnitude* to a randomly drawn donor's level, keeping the phase — filled from the noise floor, not set to zero, so no spectral hole.
3. **Finalize.** Enforce conjugate symmetry, invert the DFT, recolour through the whitener, restore the levels.

```
1 adj_ssi <- seas_adjust(unratensa, phase_rule = "zero",
2                       target_set = "bins", seed = 6376)
3 ssi_nsa <- adjusted(adj_ssi)      # classic SSI
4 seas_test(ssi_nsa)               # the same test, on the output
```

$\widehat{\Delta}$ after adjustment	-2.72
p	0.877 · fail to reject
first 12 months	left unadjusted

One honest line. The package's default branch is a different, experimental object: on a series the test classifies as a broad “band” it divides by an estimated gain and imputes nothing — and on UNRATENSA its own post-test rejects ($p \approx 3 \times 10^{-10}$). The “by construction” claim attaches to the classic call above.

SSI on UNRATENSA

Deeper Dive · not assessable

Classic SSI on UNRATENSA, 1960–2019, replayed from precomputed freqseas draws. Press **Adjust**.

whitened periodogram of Δ UNRATENSA ($\log_{10}$ power) · seasonal bins shaded, $s = 12$ · 113 targets

before $\hat{\Delta} = 122.2$ · $p < 10^{-15} \Rightarrow$ reject H_0
after ...

Adjust

Step +1

Skip to end

speed x1

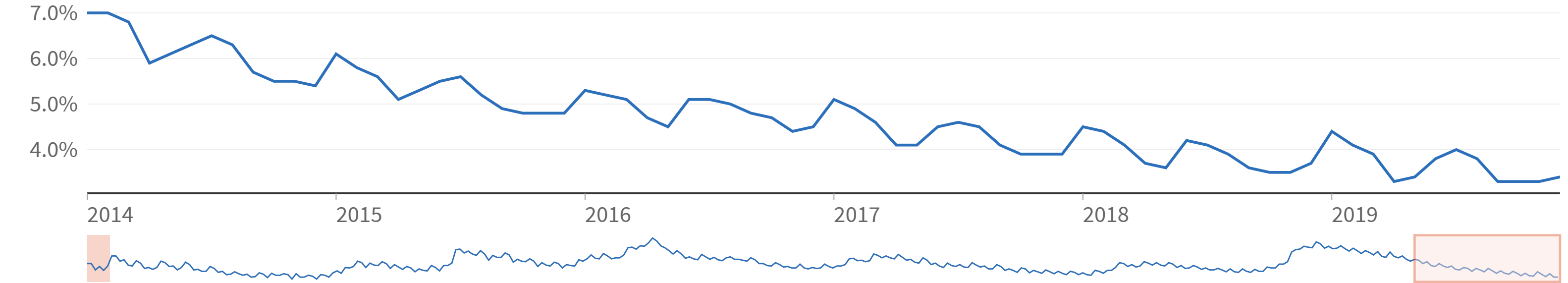
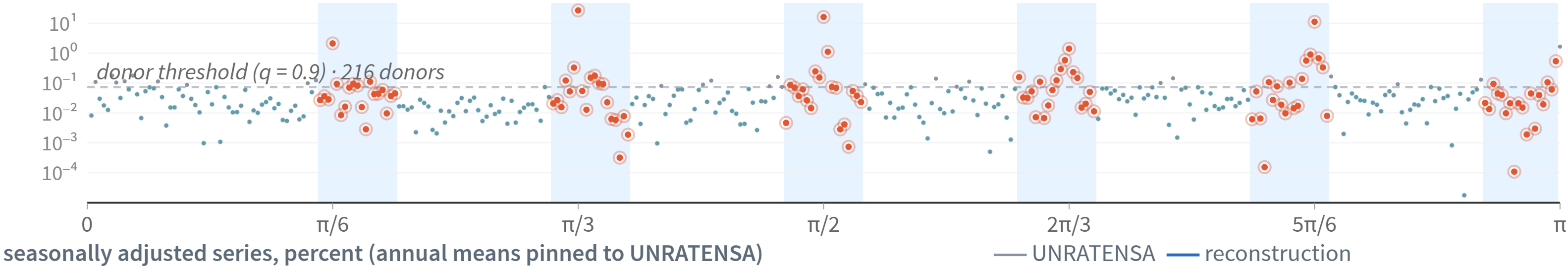
quantile

q = 0.9

 method

bootstrap

 Reseed draw 1 of 10 view: levels Reset



zoom 2014–2019 · drag the minimap window to move it · head 1960-01 to 1960-12 (12 obs) unadjusted by construction

Deeper Dive · not assessable

Epilogue: residual seasonality

Residual Seasonality in
Core Consumer Price
Inflation

Federal Reserve Board ·
2014

"The Phrase of the Day is
'Residual Seasonality'"

Wall Street Journal · 2015

U.S. says probe found
problem in seasonal
adjustment of GDP data

Reuters · 2016

Residual Seasonality in
GDP Growth Remains
after Latest BEA
Improvements

Cleveland Fed · 2019

Assessing residual
seasonality in published
outputs

UK Office for National
Statistics · 2025

Residual Seasonality in
Some Components of
PCE Inflation

Cleveland Fed · 2026

Back to the series Module 4 handed us. Its ARMA(1,2) on the published UNRATE showed **serial correlation** at the seasonal lags — a verdict on that fit. The series it was fitted to is the output of an adjustment procedure, so the course's name for that leftover is **residual seasonality**: the procedure's own serial correlation, showing up in the data it published. The seasonal ARMA terms of ARIMA $(1, 1, 2) (1, 0, 1)_{12}$ are what absorbed it.

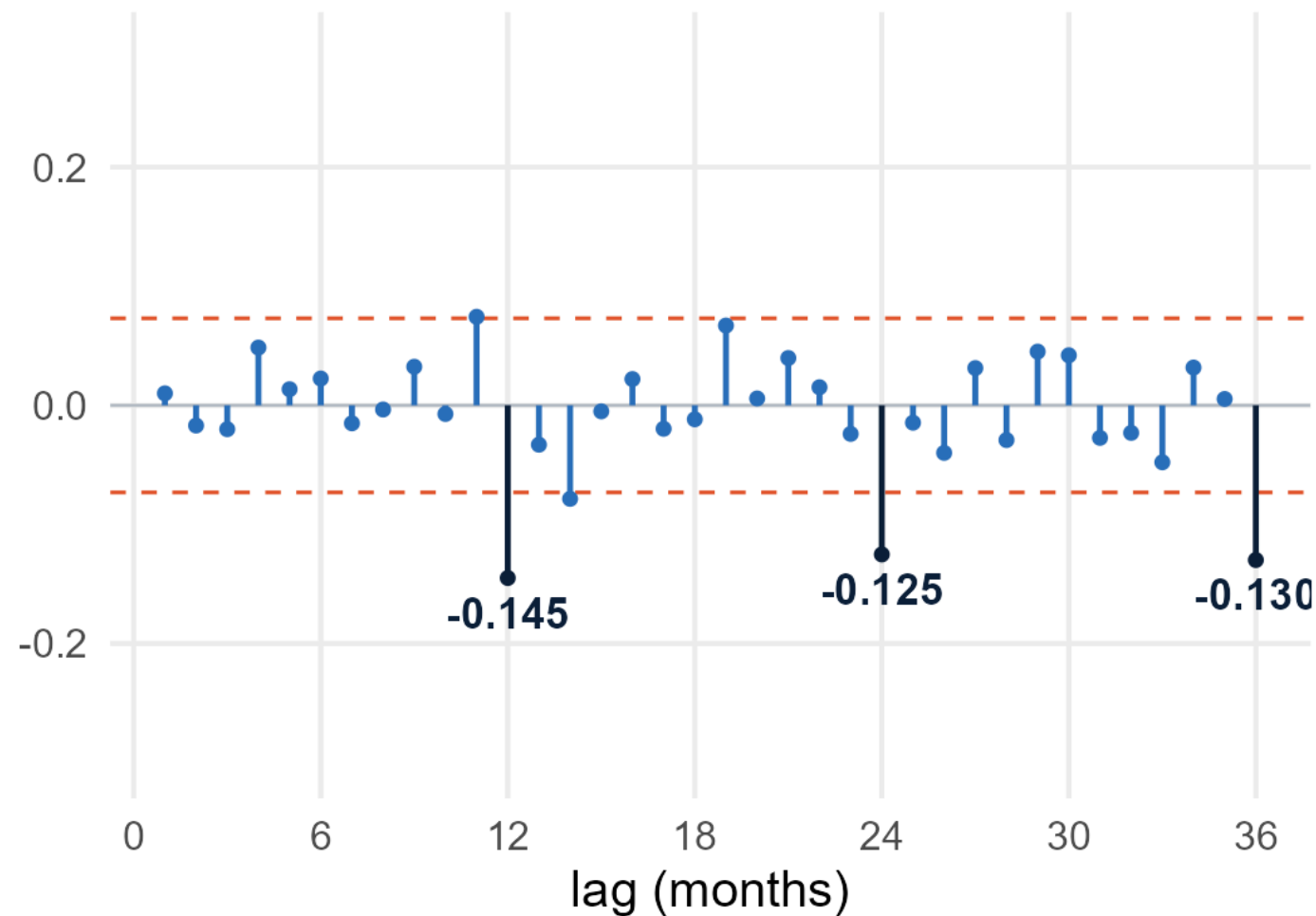
The sign says which way. $\hat{\rho}_{12} = -0.145$ and its echoes $(-0.125, -0.130)$ are *negative*. Under a non-seasonal ARMA that is consistent with the published series carrying **too little** variation at the seasonal frequencies — a seasonal **deficit**, the signature of an adjustment that removed slightly more than the seasonal component. A dip, not a peak.

Two honest limits. This is *consistent with* an adjustment-side explanation, not a claim about how the BLS runs its procedure. And the numbers do not establish *why* the dip is there.

Epilogue: before and after, on the published UNRATE

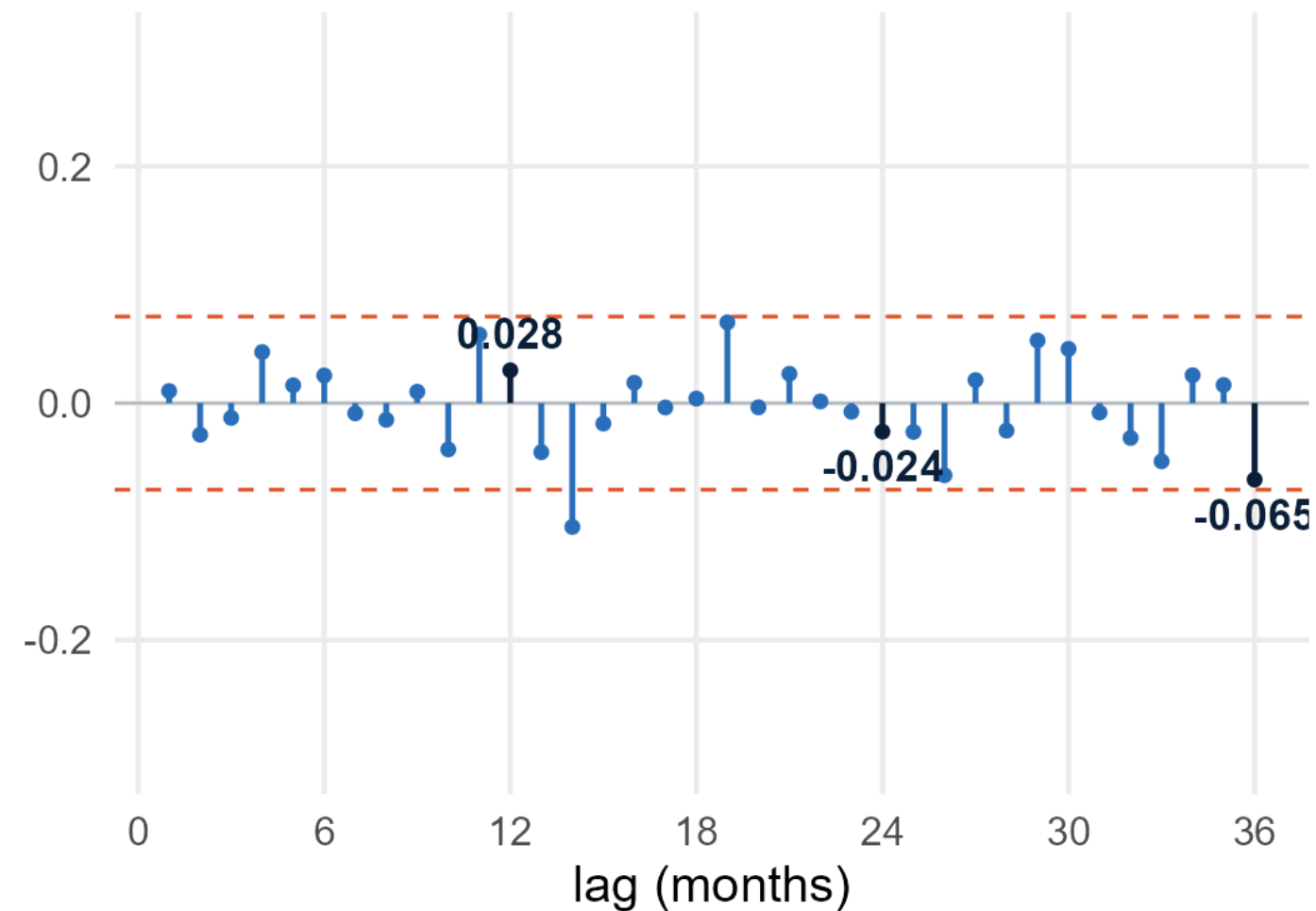
Before: ARMA(1,2) on diff(UNRATE)

negative spikes at 12, 24, 36; LB $p = 0.0012$ at $h = 24$



After: ARIMA(1,1,2)(1,0,1)[12] on UNRATE

seasonal lags inside the band; LB $p = 0.55 / 0.33$



Same series, two fits. The left picture asked Module 4's question; the right one answers it — seasonal band on, seasonal lags inside the band, Ljung-Box passing at both horizons. When someone asks what residual diagnostics are *for*, hand them this pair.

Deeper Dive · not assessable

The frequency domain agrees (notes §5.9.6): on the published series all six seasonal bins sit below their neighbours, ratio 0.75 — on the raw series the ratio is 18.4 — and $\hat{\Delta} = 0.44$, $p = 0.23$, fails to reject, because the test is one-sided for excess.

The master equation, fully lit

$$y_t = \alpha + \delta t + \sum_{j=1}^p \phi_j y_{t-j} + \sum_{l=1}^q \theta_l \epsilon_{t-l} + \epsilon_t$$

— and the seasonal block, written out, is the same structure at lag s , multiplied in:

$$\Phi_p(L) \Phi_P(L^s) (1-L)^d (1-L^s)^D y_t = \Theta_q(L) \Theta_Q(L^s) \epsilon_t$$

Knob	Module	State
$\alpha, \delta t$ — level and trend	L1, L2	on
ϕ_j, θ_l — the base band	L3, L4	on
$(1-L)^d$ — differencing at the base frequency	L2	on
$\Phi_P(L^s), \Theta_Q(L^s), (1-L^s)^D$ — the seasonal band	L5	on
ϵ_t	L1	on

The fitting toolkit built since Module 1 is complete. From here on, every module is about what you *do* with a fitted model, or about lifting it to several variables, or about letting the variance have a master equation of its own.

Key takeaways

1. **Residuals should behave like innovations.** e_t versus ϵ_t is the whole foundation of diagnostics. Ljung-Box is the omnibus check: $Q(h) \stackrel{a}{\sim} \chi^2_{h - \text{fitdf}}$, $h = 10$ and $h = 24$ on monthly data, fitdf set honestly, and a large p -value is good news.
2. **Serial correlation is the model's, not the data's.** The data has *autocorrelation*. A fitted model earns the verdict *serial correlation*. An adjustment procedure that leaves seasonal autocorrelation in a published series earns *residual seasonality*.
3. **Seasonality is a line you draw.** Four traditions, none nesting the others; whatever lands inside your definition is seasonal by construction. The one safe non-example: monthly sampling alone makes nothing seasonal.
4. **Three paths, and when each applies.** Deterministic $\rightarrow$ dummies (on UNRATENSA: fixed part absorbed, drifting part left). Stochastic $\rightarrow$ SARIMA (winner $(1, 1, 2) (0, 1, 1)_{12}$, by diagnostics and interior coefficients, because IC cannot compare across differencing orders). “Gone, not modeled” $\rightarrow$ adjustment (one X-13 call reproduces the published series). And D is a commitment the data push back on from both sides.
5. **The Module 6 handoff:** ARIMA $(1, 1, 2) (1, 0, 1)_{12}$ on the *published, adjusted* UNRATE — diagnostic-clean at both horizons, its seasonal ARMA terms absorbing the residual seasonality the adjustment left behind.

Next time: forecasting

Module 5 hands Module 6 a **diagnostic-clean univariate model** — the $ARIMA(1, 1, 2)(1, 0, 1)_{12}$ on UNRATE that passed Ljung-Box at both horizons.

You have a model. You have checked it. Now the question is what you use it for.

- **Forecasting:** the recursive computation of $\hat{y}_{t+h|t}$ — point, interval, density.
- **Forecast uncertainty:** why the interval grows with the horizon, built from the σ^2 we have tracked since Module 1.
- **Train/test splits for time series:** random splits are wrong; rolling windows are right.
- **The workflow:** split $\rightarrow$ fit on the training window $\rightarrow$ forecast $\rightarrow$ evaluate on the test window. Today was fit + check. Modules 6 and 7 add the rest.

A forecast is a conditional mean plus honest uncertainty; classical ARMA gives you both.